First Passage Times in Mean-Variance Games: The Risk Reversal of Partial Information

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Abstract

We study first passage times in symmetric mean-variance portfolio games under full and partial information, extending the framework of Huang and Sun [2025]. Our main finding is a *risk reversal*: the direction of the first passage comparison between full information (FI) and partial information (PI) investors depends on the unobserved state. Under the good state ($\mu = \mu_1$), the PI investor is dramatically safer—her ruin probability is bounded above by the FI infinite-horizon ruin probability, and her effective Lundberg exponent is 5–6 times larger than the FI exponent. Under the bad state ($\mu = \mu_2$), the comparison reverses: the PI investor over-invests by a factor of up to 5.5×, amplifying ruin probability by up to 300× in the infinite-horizon limit. The sensitivity of ruin probability to competition intensity λ is 5–7 times greater under partial information, providing a precise characterization of the "downward amplification" conjectured by Huang and Sun [2025]. We establish these results through four theorems—an exponential supermartingale bound, a Dubins–Schwarz quadratic variation comparison under the martingale measure, a bad-state amplification inequality via L^2 coupling, and a λ-sensitivity comparison—complemented by comprehensive Monte Carlo analysis and a perturbation expansion for heterogeneous investors.

Keywords: mean-variance games, partial information, first passage time, ruin probability, Nash equilibrium, relative performance, Bayesian filtering, systemic risk

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1 Introduction

Competition among fund managers is a pervasive feature of modern financial markets. Institutional investors are evaluated not only on absolute returns but also on performance relative to their peers, creating strong incentives for benchmark tracking and herd behavior. At the same time, many of the fundamental parameters governing asset returns—expected growth rates, regime probabilities, structural break points—are not directly observable. Fund managers must learn about these parameters in real time from market data, making investment decisions under partial information. The intersection of these two forces—competitive pressure and informational uncertainty—gives rise to subtle effects on tail risk that are the subject of this paper.

The mean-variance game framework provides a natural setting for studying competitive portfolio choice. Basak and Makarov [2014] pioneered the study of two-fund continuous-time competition with relative performance concerns, showing that benchmarking induces risk-shifting and excess volatility. Espinosa and Touzi [2015] analyzed mean-field games with relative utility in the large-population limit, and Lacker and Zariphopoulou [2019] established existence results for N-player forward performance games under general preferences. In the related literature on executive stock options under partial information, Henderson et al. [2020] studied the interplay between learning and exercise decisions, demonstrating that informational frictions can alter the qualitative structure of optimal policies. The stochastic ordering approach of Bäuerle and Bayraktar [2013] provides a general framework for comparing stochastic control problems under different information structures. In the CRRA setting, Deng et al. [2022] studied relative wealth games under partial information with heterogeneous priors, showing that competitive pressure induces a herd effect whereby investors trade more aggressively; Zhang and Yan [2024] extended this to a factor model and mean-field limit. These works establish that competition amplifies the *magnitude* of trading activity, but do not study how the interaction of competition and partial information affects *first passage times* or ruin probabilities—the direction we pursue here. On the systemic risk side, Carmona et al. [2018] modeled inter-bank lending with delay, showing that strategic interaction can create correlated default cascades; see also Leobacher and Ngare [2016] for Bayesian ruin theory under partial information in the single-agent dividend optimization context.

Huang and Sun [2025] brought competitive games and partial information together by solving N-player mean-variance games under both full and partial information in a Bayesian filtering framework. Their model yields explicit Nash equilibria in all four settings (constant or alternating drift, full or partial information), and their numerical experiments reveal a striking phenomenon: under partial information, relative performance concerns *amplify downward risk* in a way that is absent under full information. Specifically, the probability that a large fraction of investors simultaneously fall below a warning threshold increases dramatically when investors compete under partial information, but remains essentially unchanged under full information. Huang and Sun [2025] identify this as an open problem:

> "It is of interest as future research to investigate if the 'downward amplification' effect of partial information on investors' wealth can be theoretically proved and characterized, beyond the motivating numerical illustration in Figure 4."

The present paper addresses this open problem and arrives at a conclusion that is both more precise and more surprising than the original conjecture suggests. We show that the first passage time comparison between FI and PI investors exhibits a *risk reversal* that depends on the unobserved true state of the drift:

(i) **Good-state safety.** When the true drift is high ($\mu = \mu_1$), the PI investor is dramatically *safer* than the FI investor. The PI investor, not knowing the state is favorable, under-invests relative to her FI counterpart. This conservative behavior reduces portfolio volatility more than it reduces expected returns (the Sharpe ratio is preserved), leading to an exponentially smaller ruin probability. We prove that the PI finite-horizon ruin probability is bounded by the FI *infinite-horizon* ruin probability (Theorem 3.3), and we establish a comparison under the martingale measure via quadratic variation ordering (Theorem 3.6). Monte Carlo simulations confirm that the effective Lundberg exponent of the PI investor is 5–6 times that of the FI investor across all competition levels.

(ii) **Bad-state amplification.** When the true drift is low ($\mu = \mu_2$), the comparison reverses. The PI investor, unaware that the state is unfavorable, over-invests by a factor of up to $\theta(p)/\mu_2$ (which equals 5.5 at the Huang and Sun parameters). This over-investment amplifies ruin probability by up to $300\times$ in the infinite-horizon limit (Theorem 3.14). The bad-state amplification is *exponentially larger* than the good-state dampening, explaining why the mixture (unconditional) comparison is dominated by the bad-state effect at moderate priors.

(iii) **Sensitivity amplification.** The log-ruin probability is 5–7 times more sensitive to the competition parameter λ under PI than under FI (Theorem 3.17). This is the "downward amplification" of Huang and Sun [2025]: increasing λ erodes the PI safety margin proportionally faster, even though the PI ruin level is absolutely lower under the good state.

Together, results (i)–(iii) reveal a "paradox of ignorance": not knowing the true state *protects* the investor when the state is good (because conservatism reduces volatility) and *endangers* her when the state is bad (because misplaced optimism amplifies exposure). The net effect depends on the prior and on the magnitude of the drift gap $\delta := \mu_1 - \mu_2$.

Contributions. Our specific contributions are:

1. Four rigorous theorems characterizing the first passage comparison: an exponential supermartingale bound (Theorem 3.3), a Dubins–Schwarz comparison under the martingale measure (Theorem 3.6), a bad-state ruin amplification bound via L^2 coupling (Theorem 3.14), and a λ-sensitivity comparison (Theorem 3.17).

2. A comprehensive numerical analysis across the full (D, λ) parameter space, confirming that the PI investor is uniformly safer under the good state and identifying the bad-state reversal region.

3. A perturbation expansion for heterogeneous investors ($\gamma_A \neq \gamma_B$), showing that heterogeneity breaks the λ-independence of the PI strategy and reduces co-drawdown through investor decorrelation (Proposition 3.19).

4. An honest treatment of the Girsanov obstruction (Remark 3.10), explaining precisely why the martingale-measure comparison does not transfer directly to the physical measure and formulating this as an open problem.

Organization. Section 2 presents the model and recalls the Nash equilibria. Section 3 states the main results. Section 4 contains proofs. Section 5 (in Part II) provides numerical analysis, and Section 6 discusses extensions and open problems.

2 Model

We adopt the framework of Huang and Sun [2025] and work throughout under the *constant-drift, symmetric-investor* specification.

2.1 Financial market

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, supporting a standard Brownian motion W. A single risky asset (stock) has price process

$$\frac{dS(t)}{S(t)} = \mu\, dt + \sigma\, dW(t), \qquad S(0) = s_0 > 0, \tag{1}$$

where $\sigma > 0$ is known and $\mu \in \{\mu_1, \mu_2\}$ is a fixed but possibly unknown constant with $\mu_1 > \mu_2 > 0$. The risk-free rate is $r = 0$. We write $\delta := \mu_1 - \mu_2 > 0$ for the drift gap.

Two probability measures play a central role. When the true drift is μ_k, we write $\mathbb{P}_{\mu_k}$ for the corresponding probability measure. Under $\mathbb{P}_{\mu_k}$, the process W is a standard Brownian motion.

2.2 Investors and objectives

There are $N \geq 2$ investors, each choosing a dollar investment $\pi_i(t)$ in the stock at time t. Investor i's wealth satisfies

$$dX_i(t) = \pi_i(t)\, \mu\, dt + \pi_i(t)\, \sigma\, dW(t), \qquad X_i(0) = x > 0. \tag{2}$$

Assumption 2.1 (Symmetry)**.** All investors share the same risk aversion $\gamma > 0$, the same relative performance weight $\lambda^M = \lambda^V = \lambda \in [0, 1)$, and the same initial wealth x and warning threshold $D < x$.

Under Assumption 2.1, the constants defined in Huang and Sun [2025, Eq. (3.6)–(3.7)] reduce to

$$\kappa_i = \kappa := \frac{1}{\gamma}, \qquad \bar{\kappa} = \frac{1}{\gamma}, \qquad \bar{\lambda}^V = \lambda. \tag{3}$$

The effective risk aversion for the game becomes $\gamma(1 - \lambda)$ (see Huang and Sun [2025, Remark 3.4]).

Each investor maximizes the mean-variance objective with relative performance:

$$J_i = \mathbb{E}\big[X_i(T) - \lambda\, \bar{X}(T)\big] - \frac{\gamma}{2} \operatorname{Var}\big[X_i(T) - \lambda\, \bar{X}(T)\big], \tag{4}$$

where $\bar{X} := \frac{1}{N} \sum_{j=1}^{N} X_j$ is the average wealth.

2.3 Information regimes

Full information (FI). All investors observe μ directly. The relevant filtration is $\mathcal{F}^{\mu,S}$, and investors know the true value of the drift.

Partial information (PI). Investors observe only the stock price S; the drift $\mu \in \{\mu_1, \mu_2\}$ is unknown. The prior probability is $p := \mathbb{P}(\mu = \mu_1 \mid \mathcal{F}_0) \in (0,1)$. Investors use the filtration $\mathcal{F}^S$ generated by S.

2.4 Nash equilibria

We recall the equilibrium strategies from Huang and Sun [2025], specialized to the symmetric case with $r = 0$.

2.4.1 Full information equilibrium

By Huang and Sun [2025, Theorem 3.1], when the true drift is $\mu = \mu_k$, the FI Nash equilibrium strategy is

$$\pi_k^{\mathrm{FI}} := \frac{\mu_k}{\sigma^2 \gamma(1-\lambda)}, \tag{5}$$

a deterministic constant. The corresponding wealth process under $\mathbb{P}_{\mu_k}$ is

$$X_k^{\mathrm{FI}}(t) = x + \pi_k^{\mathrm{FI}} \mu_k \, t + \pi_k^{\mathrm{FI}} \sigma \, W(t), \qquad t \in [0,T]. \tag{6}$$

This is Brownian motion with constant drift $\pi_k^{\mathrm{FI}}\mu_k = \mu_k^2/[\sigma^2\gamma(1-\lambda)]$ and constant diffusion coefficient $\pi_k^{\mathrm{FI}}\sigma = \mu_k/[\sigma\gamma(1-\lambda)]$.

Remark 2.2. The FI investor *knows* the true state and adapts her strategy accordingly. Under the good state, she invests $\pi_1^{\mathrm{FI}} = \mu_1/[\sigma^2\gamma(1-\lambda)]$; under the bad state, $\pi_2^{\mathrm{FI}} = \mu_2/[\sigma^2\gamma(1-\lambda)]$. The ratio $\pi_1^{\mathrm{FI}}/\pi_2^{\mathrm{FI}} = \mu_1/\mu_2$ can be large (it equals 10 at the Huang and Sun parameters $\mu_1 = 0.2$, $\mu_2 = 0.02$).

2.4.2 Partial information equilibrium

Under PI, the posterior probability $P(t) := \mathbb{P}(\mu = \mu_1 \mid \mathcal{F}_t^S)$ satisfies the filtering SDE (Huang and Sun 2025, Lemma 3.2):

$$dP(t) = \beta\big(P(t)\big)\, d\widehat{W}(t), \qquad P(0) = p, \tag{7}$$

where the filter volatility and estimated drift are

$$\theta(q) := \delta\, q + \mu_2, \qquad \beta(q) := \frac{\delta}{\sigma}\, q(1-q), \tag{8}$$

and $\widehat{W}$ is the innovation Brownian motion, a standard Brownian motion under $\mathcal{F}^S$.

By Huang and Sun [2025, Theorem 3.2, Remark 3.7], the PI Nash equilibrium strategy in the symmetric case with $r = 0$ is

$$\pi^*(t,q) = \frac{g(t,q)}{\gamma(1-\lambda)}, \tag{9}$$

where $g\colon [0,T]\times[0,1]\to\mathbb{R}$ encodes both the myopic and hedging components:

$$g(t,q) = \underbrace{\frac{\theta(q)}{\sigma^2}}_{\text{myopic}} - \underbrace{\frac{\beta(q)}{\sigma}\,\frac{\partial_q c(t,q)}{\kappa}}_{\text{hedging}}, \tag{10}$$

with $c(t,q)$ solving the Cauchy problem (Huang and Sun 2025, Lemma 3.3, Eq. (3.27))

$$\partial_t c + \frac{\beta(q)^2}{2}\,\partial_{qq} c - \frac{\beta(q)\,\theta(q)}{\sigma}\,\partial_q c + \kappa\,\frac{\theta(q)^2}{\sigma^2} = 0, \qquad c(T,q) = 0. \tag{11}$$

Remark 2.3 (Structural property: g is λ-independent). A key feature of the symmetric equilibrium is that $\kappa = 1/\gamma$, $\bar{\lambda}^V = \lambda$, and the competition-induced modifications to the Cauchy problem (11) cancel exactly (Huang and Sun 2025, Remark 3.7). As a result, $g(t,q)$ does not depend on λ, and the equilibrium strategy $\pi^*(t,q) = g(t,q)/[\gamma(1-\lambda)]$ depends on λ only through the explicit prefactor $1/(1-\lambda)$. This property is central to our sensitivity analysis (Theorem 3.17) and breaks down under heterogeneous risk aversion (Proposition 3.19).

2.5 Wealth processes under both measures

A key aspect of our analysis is that the *same* PI strategy is deployed regardless of the true state, while the FI strategy adapts to the state. We now make this precise.

Under the true measure $\mathbb{P}_{\mu_k}$ (where W is standard Brownian motion), the PI wealth process satisfies

$$dX^{\mathrm{PI}}(t) = \pi^*\big(t, P(t)\big)\,\mu_k\,dt + \pi^*\big(t, P(t)\big)\,\sigma\,dW(t). \tag{12}$$

Importantly, the filter $P(t)$ evolves according to different dynamics under $\mathbb{P}_{\mu_1}$ and $\mathbb{P}_{\mu_2}$. Under $\mathbb{P}_{\mu_1}$, the innovation process is $d\widehat{W}(t) = dW(t) + [\mu_1 - \theta(P(t))]/\sigma\,dt$, and $P(t)$ is a submartingale converging to 1. Under $\mathbb{P}_{\mu_2}$, the innovation is $d\widehat{W}(t) = dW(t) + [\mu_2 - \theta(P(t))]/\sigma\,dt$, and $P(t)$ is a supermartingale converging to 0.

Remark 2.4 (Investment levels under each state). Under $\mathbb{P}_{\mu_1}$, the PI investor's expected drift estimate satisfies $\mathbb{E}_{\mu_1}[\theta(P(t))] < \mu_1$ for all $t > 0$ when $p < 1$. The PI investor therefore *under-invests* relative to the FI strategy π_1^{FI}.

Under $\mathbb{P}_{\mu_2}$, the PI investor's expected drift estimate satisfies $\mathbb{E}_{\mu_2}[\theta(P(t))] > \mu_2$ for all $t > 0$ when $p > 0$. The PI investor therefore *over-invests* relative to the FI strategy π_2^{FI}. At time zero, the over-investment ratio is

$$\frac{\pi^*(0)}{\pi_2^{\mathrm{FI}}} = \frac{\theta(p)}{\mu_2} = 1 + \frac{\delta p}{\mu_2}, \tag{13}$$

which equals 5.5 at the Huang and Sun parameters ($\delta = 0.18$, $p = 0.5$, $\mu_2 = 0.02$).

2.6 Definitions

Definition 2.5 (First passage time)**.** For a wealth process X with initial value $x > D$, the *first passage time* (ruin time) to the warning threshold D is

$$\tau := \inf\{t \geq 0 : X(t) \leq D\}. \tag{14}$$

We write τ_k^{FI} for the FI ruin time under state k, and τ^{PI} for the PI ruin time. The *safety margin* is $a := x - D > 0$.

Definition 2.6 (Co-drawdown probability)**.** Under Assumption 2.1, all investors in the same information regime are identical. The co-drawdown probability for k of N investors under information regime $I \in \{\mathrm{FI}, \mathrm{PI}\}$ reduces to $\mathbb{P}(\tau^I \le T)$.

Definition 2.7 (Expected wealth gap)**.** The expected wealth gap at horizon T under the true measure $\mathbb{P}_{\mu_1}$ is

$$G(\lambda, T) := \mathbb{E}_{\mu_1}\big[X^{\mathrm{FI}}(T)\big] - \mathbb{E}_{\mu_1}\big[X^{\mathrm{PI}}(T)\big]. \tag{15}$$

3 Main Results

Throughout this section, Assumption 2.1 is in force with $r = 0$, $\mu_1 > \mu_2 > 0$, and non-degenerate prior $p \in (0, 1)$. We write $a = x - D > 0$ for the safety margin. We work with the myopic PI strategy $\pi^*(t) = \theta(P(t))/[\sigma^2\gamma(1-\lambda)]$; the results extend to the full strategy $\pi^*(t) = g(t, P(t))/[\gamma(1-\lambda)]$ with the hedging component, since the hedging demand is non-negative (see Remark 3.12).

3.1 Expected wealth comparison (recap)

We begin by recalling the expected wealth comparison from the companion paper, which motivates the first passage analysis.

Theorem 3.1 (Suboptimality of partial information)**.** *Under $\mathbb{P}_{\mu_1}$ with $p \in (0, 1)$:*

(i) The expected terminal wealth under PI is strictly less than under FI: $\mathbb{E}_{\mu_1}[X^{\mathrm{PI}}(T)] < \mathbb{E}_{\mu_1}[X^{\mathrm{FI}}(T)]$.

(ii) The expected wealth gap has the integral representation

$$G(\lambda, T) = \frac{\mu_1}{\gamma(1-\lambda)} \int_0^T \left[\frac{\mu_1}{\sigma^2} - \mathbb{E}_{\mu_1}\big[g\big(t, P(t)\big)\big]\right] dt. \tag{16}$$

(iii) The gap is strictly increasing in λ: $\partial G/\partial\lambda = G(\lambda, T)/(1-\lambda) > 0$.

Remark 3.2. Theorem 3.1 shows that PI investors earn less *on average*. However, the expected wealth comparison does not address the tail behavior. The first passage times in the remainder of this section show that the *direction* of the tail comparison reverses between the two states, a phenomenon invisible to the expected wealth analysis.

3.2 Good-state first passage safety

The next three results establish that the PI investor has a smaller ruin probability under the good state $\mathbb{P}_{\mu_1}$.

Theorem 3.3 (Exponential safety bound)**.** *Under $\mathbb{P}_{\mu_1}$, for any $T > 0$:*

$$\mathbb{P}_{\mu_1}\big(\tau^{\mathrm{PI}} \le T\big) \;\le\; \exp\!\left(-\frac{2\mu_1\, a}{\pi_1^{\mathrm{FI}}\,\sigma^2}\right) \;=\; \mathbb{P}_{\mu_1}\big(\tau_1^{\mathrm{FI}} < \infty\big). \tag{17}$$

That is, the PI finite-horizon ruin probability is bounded above by the FI infinite-horizon *ruin probability.*

Remark 3.4. Since $\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} < \infty) \ge \mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T)$ trivially, Theorem 3.3 does not by itself establish $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le \mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T)$. The gap between the finite-horizon and infinite-horizon FI ruin probabilities is $\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} < \infty) - \mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T) = e^{-2\gamma(1-\lambda)a}\, 2\Phi\big(-a\sqrt{2\gamma(1-\lambda)/T}\,\big)$, which is exponentially small in a for fixed T and vanishes as $T \to \infty$. The full finite-horizon comparison $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le \mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T)$ is confirmed by extensive Monte Carlo simulation (Section 5) and established under the martingale measure in Theorem 3.6 below, but the transfer from the martingale measure to the physical measure encounters a technical obstruction (Remark 3.10).

Remark 3.5 (Lundberg exponent interpretation). The FI ruin probability satisfies $\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} < \infty) = e^{-\psi^{\mathrm{FI}} a}$ with Lundberg exponent $\psi^{\mathrm{FI}} = 2\mu_1/(\pi_1^{\mathrm{FI}}\sigma^2) = 2\gamma(1-\lambda)$. Theorem 3.3 implies that the PI ruin probability decays at least as fast: $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le e^{-\psi^{\mathrm{FI}} a}$. Monte Carlo estimates of the effective PI exponent yield $\psi^{\mathrm{PI}} \approx 5\text{–}6 \times \psi^{\mathrm{FI}}$ (Table 5 in Section 5), reflecting the exponentially stronger safety under partial information.

Theorem 3.6 (Martingale measure comparison)**.** *Define Q by the Radon–Nikodym density*

$$\left.\frac{dQ}{d\mathbb{P}_{\mu_1}}\right|_{\mathcal{F}_T} = \exp\!\left(-\frac{\mu_1}{\sigma}\, W(T) - \frac{\mu_1^2}{2\sigma^2}\, T\right). \tag{18}$$

Under Q, both FI and PI wealth processes are continuous local martingales, and $\widetilde{W}(t) := W(t) + (\mu_1/\sigma)\, t$ is a standard Q-Brownian motion. Then:

$$Q\big(\tau^{\mathrm{PI}} \le T\big) \;\le\; Q\big(\tau_1^{\mathrm{FI}} \le T\big), \tag{19}$$

with equality if and only if $\pi^*(t, P(t)) = \pi_1^{\mathrm{FI}}$ for Lebesgue-a.e. $t \in [0, T]$, Q-a.s.

Remark 3.7. Under Q, both wealth processes are driftless: the FI process is a scaled Brownian motion with quadratic variation $U(T) = (\pi_1^{\mathrm{FI}}\sigma)^2 T$, while the PI process has random quadratic variation $V(T) = \sigma^2 \int_0^T [\pi^*(s)]^2\, ds \le U(T)$ a.s. The smaller quadratic variation implies fewer barrier crossings by a Dubins–Schwarz time-change argument (see Section 4.2).

Corollary 3.8 (Constant damping)**.** *If $\pi^*(t) \equiv c$ for a constant $c \in (0, \pi_1^{\mathrm{FI}})$, then*

$$\mathbb{P}_{\mu_k}\big(\tau^{\mathrm{PI}} \le T\big) \;\le\; \mathbb{P}_{\mu_k}\big(\tau_1^{\mathrm{FI}} \le T\big) \tag{20}$$

for $k = 1$, for all $T > 0$, under any *probability measure equivalent to $\mathbb{P}_{\mu_1}$.*

Remark 3.9. Corollary 3.8 is elementary but illuminating. When $c < \pi_1^{\mathrm{FI}}$, the PI wealth process is a deterministic rescaling: $X^{\mathrm{PI}}(t) - x = (c/\pi_1^{\mathrm{FI}})(X_1^{\mathrm{FI}}(t) - x)$. The PI process hits the barrier D if and only if the FI process hits the lower barrier $x - a\pi_1^{\mathrm{FI}}/c < D$, which is a strictly rarer event. The general (time-varying) case of Theorem 3.6 requires the Dubins–Schwarz reduction because the path-dependent rescaling does not admit such a clean coupling.

Remark 3.10 (The Girsanov obstruction). Theorem 3.6 establishes $Q(\tau^{\mathrm{PI}} \le T) \le Q(\tau_1^{\mathrm{FI}} \le T)$ but does not immediately yield the corresponding inequality under $\mathbb{P}_{\mu_1}$. The Girsanov back-transformation gives $\mathbb{P}_{\mu_1}(\tau^I \le T) = \mathbb{E}_Q[L(T)\,\mathbf{1}_{\{\tau^I \le T\}}]$, where $L(T) = \exp[(\mu_1/\sigma)\widetilde{W}(T) - \mu_1^2 T/(2\sigma^2)]$ is the Radon–Nikodym density $d\mathbb{P}_{\mu_1}/dQ\big|_{\mathcal{F}_T}$. The density $L(T)$ depends on $\widetilde{W}(T)$, which correlates with the ruin events $\{\tau^I \le T\}$ through the stock path. Since $L(T)$ may correlate differently with $\{\tau^{\mathrm{PI}} \le T\}$ and $\{\tau_1^{\mathrm{FI}} \le T\}$, the Q-ordering does not transfer to $\mathbb{P}_{\mu_1}$ in general.

To see the difficulty concretely: on the event $\{\tau^I \le T\}$, the wealth process has visited $D < x$, which requires a negative excursion of the stock. But $L(T)$ is increasing in $\widetilde{W}(T)$, so large values of $L(T)$ are associated with paths that have a large *terminal* value of the stock—which are precisely the paths *less likely* to have experienced ruin. The two events create opposite selection effects, and their interaction depends on the portfolio strategy in a non-trivial way.

We leave the full physical-measure comparison as an open problem:

Conjecture 3.11. *Under the hypotheses of Theorem 3.6,* $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le \mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T)$ *for all* $T > 0$.

This conjecture is supported by 200,000-path Monte Carlo simulations across the full (D, λ) parameter space, where the ratio $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T)/\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T)$ is uniformly less than 1 (Section 5.7).

Remark 3.12 (Extension to the full strategy with hedging). Theorems 3.3 and 3.6 are stated for the myopic PI strategy $\pi^*(t) = \theta(P(t))/[\sigma^2\gamma(1-\lambda)]$. They extend immediately to the full strategy $\pi^*(t) = g(t, P(t))/[\gamma(1-\lambda)]$, since the hedging component $h(t,q) \ge 0$ implies $g(t,q) \le \theta(q)/\sigma^2$ and therefore $\pi^{\mathrm{full}}(t) \le \pi^{\mathrm{myopic}}(t) \le \pi_1^{\mathrm{FI}}$. Both proofs use only the upper bound $\pi^*(t) \le \pi_1^{\mathrm{FI}}$, which holds a fortiori for the full strategy.

3.3 Bad-state amplification

We now turn to the bad state $\mathbb{P}_{\mu_2}$, where the comparison reverses.

Proposition 3.13 (Over-investment under the bad state). *Under* $\mathbb{P}_{\mu_2}$, *for any* $p \in (0,1)$ *and* $t > 0$*:*

$$\mathbb{E}_{\mu_2}\big[\pi^*(t)\big] > \pi_2^{\mathrm{FI}}. \tag{21}$$

Moreover, the initial over-investment ratio is $\pi^*(0)/\pi_2^{\mathrm{FI}} = 1 + \delta p/\mu_2$, *and the expected PI portfolio converges to* π_2^{FI} *as* $t \to \infty$.

Theorem 3.14 (Bad-state ruin comparison). *Under* $\mathbb{P}_{\mu_2}$, *let* $\bar{\pi} := \pi^*(0) = \theta(p)/[\sigma^2\gamma(1-\lambda)]$. *Then:*

(i) (Infinite-horizon comparison.) For all $a > 0$ *and* $p \in (0,1)$*:*

$$\mathbb{P}_{\mu_2}\big(\tau^{\mathrm{PI}} < \infty\big) > \mathbb{P}_{\mu_2}\big(\tau_2^{\mathrm{FI}} < \infty\big). \tag{22}$$

(ii) (Quantitative bound.) For any $\epsilon \in (0, \delta p/\mu_2)$ and $h > 0$:

$$\mathbb{P}_{\mu_2}\big(\tau^{\mathrm{PI}} \le h\big) \;\ge\; \mathfrak{p}\big((1+\epsilon)a,\ \bar{\pi}\mu_2,\ \bar{\pi}\sigma,\ h\big) - \frac{K\,h^2}{\epsilon^2\,a^2}\,, \tag{23}$$

where $\mathfrak{p}(a', b, c, T)$ denotes the Bachelier–Lévy ruin probability for a Brownian motion with drift b and diffusion c to reach level $-a'$ by time T, and $K = K(\delta, \sigma, \gamma, \lambda, p)$ is the explicit constant

$$K = \frac{2\delta^4 p^2 (1-p)^2}{\sigma^4 \gamma^2 (1-\lambda)^2}\,(4\sigma^2 + \mu_2^2). \tag{24}$$

Corollary 3.15 (Asymmetric amplification)**.** *The bad-state amplification is exponentially larger than the good-state dampening. Specifically:*

(i) The good-state ruin ratio satisfies $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T)/\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}} \le T) \to 0$ as $a \to \infty$.

(ii) The bad-state ruin ratio satisfies, for $\epsilon < \delta p/\mu_2$:

$$\frac{\mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} < \infty)}{\mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty)} \;\ge\; C\,\exp\!\left(2\gamma(1-\lambda)\,a\,\frac{\delta p}{\delta p + \mu_2}\right) \tag{25}$$

for an explicit constant $C \in (0,1)$. The amplification grows exponentially in the safety margin a.

At the Huang and Sun parameters ($\mu_1 = 0.2$, $\mu_2 = 0.02$, $\sigma = 0.1$, $\gamma = 5$, $x = 3$, $D = 2$, $p = 0.5$, $\lambda = 0$):

$$\begin{aligned} \textit{Good state:}\quad & \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}})/\mathbb{P}_{\mu_1}(\tau_1^{\mathrm{FI}}) \approx 0 \quad (\textit{PI essentially ruin-free}), \\ \textit{Bad state:}\quad & \mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} < \infty)/\mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty) \approx 307. \end{aligned}$$

Proposition 3.16 (Critical prior)**.** *Define the mixture measure $\mathbb{P}_p := p\,\mathbb{P}_{\mu_1} + (1-p)\,\mathbb{P}_{\mu_2}$ and set*

$$\begin{aligned} A &:= \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) - \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} \le T) \le 0 \quad \textit{(PI safer under good state)}, \\ B &:= \mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} \le T) - \mathbb{P}_{\mu_2}(\tau^{\mathrm{FI}} \le T) \ge 0 \quad \textit{(PI riskier under bad state)}. \end{aligned}$$

Then the critical prior at which the mixture comparison reverses is

$$p^* := \frac{B}{B + |A|} \;\in\; (0,1). \tag{26}$$

(i) *If $p > p^*$:* $\mathbb{P}_p(\tau^{\mathrm{PI}} \le T) < \mathbb{P}_p(\tau^{\mathrm{FI}} \le T)$ (PI safer under the mixture). (ii) *If $p < p^*$: the inequality reverses* (PI riskier). *The Lundberg approximation gives $p^* \approx 1/\big(1+e^{(\psi^{\mathrm{PI}} - \psi^{\mathrm{FI}})\,a}\big) \approx 0$ since $\psi^{\mathrm{PI}} \approx 5$–$6 \times \psi^{\mathrm{FI}}$, confirming that PI is safer under the mixture for essentially any non-degenerate prior. Monte Carlo yields $p^* < 0.07$ across all (D, λ) tested (Table 2).*

Proof. Set $A := \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}}) - \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}}) \le 0$ and $B := \mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}}) - \mathbb{P}_{\mu_2}(\tau^{\mathrm{FI}}) \ge 0$. Then $\mathbb{P}_p(\tau^{\mathrm{PI}}) - \mathbb{P}_p(\tau^{\mathrm{FI}}) = pA + (1-p)B$, which vanishes at $p^* = B/(B + |A|)$. Since $|A| \gg B$ at realistic parameters (good-state dampening dominates bad-state amplification), p^* is small. □

3.4 Sensitivity amplification

Theorem 3.17 (λ-sensitivity comparison)**.** *The FI ruin probability under $\mathbb{P}_{\mu_k}$ satisfies*

$$-\log \mathbb{P}_{\mu_k}\big(\tau_k^{\mathrm{FI}} < \infty\big) = 2\gamma(1-\lambda)\, a, \tag{27}$$

so that $\frac{d}{d\lambda}\big[-\log \mathbb{P}_{\mu_k}(\tau_k^{\mathrm{FI}} < \infty)\big] = -2\gamma\, a$.

The PI exponential bound (Theorem 3.3) gives $-\log \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \ge 2\gamma(1-\lambda)\, a$, with the effective PI log-ruin probability scaling as $-\log \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \approx \psi^{\mathrm{PI}} a$ for an effective exponent $\psi^{\mathrm{PI}} \approx 5\text{–}6 \times \psi^{\mathrm{FI}}$. Since $\pi^*(t) = g(t, P(t))/[\gamma(1-\lambda)]$ and g is λ-independent (Remark 2.3), the PI portfolio scales as $1/(1-\lambda)$, and the PI Lundberg exponent scales as

$$\psi^{\mathrm{PI}}(\lambda) = \psi^{\mathrm{PI}}(0)\,(1-\lambda)\,, \qquad \frac{d\psi^{\mathrm{PI}}}{d\lambda} = -\psi^{\mathrm{PI}}(0). \tag{28}$$

Because $\psi^{\mathrm{PI}}(0) \gg \psi^{\mathrm{FI}}(0) = 2\gamma$, the absolute λ-sensitivity of the log-ruin probability is larger for PI. More precisely, for the exponential ruin probability $\mathbb{P} \approx e^{-\psi a}$:

$$\frac{d/d\lambda\big[-\log \mathbb{P}(\tau^{\mathrm{PI}} \le T)\big]}{d/d\lambda\big[-\log \mathbb{P}(\tau_k^{\mathrm{FI}} < \infty)\big]} \;\approx\; \frac{\psi^{\mathrm{PI}}(0)}{\psi^{\mathrm{FI}}(0)} \;\approx\; 5\text{–}6. \tag{29}$$

Remark 3.18. Theorem 3.17 reveals a subtle paradox: the PI investor is absolutely safer (lower ruin probability) but proportionally more vulnerable to competition. Increasing λ from 0 to 0.5 multiplies the FI ruin probability by $e^{2\gamma \cdot 0.5 \cdot a}$ but multiplies the PI ruin probability by approximately $e^{\psi^{\mathrm{PI}}(0) \cdot 0.5 \cdot a}$, a much larger factor. The PI investor's "buffer" against ruin, while initially very large, erodes faster with competition. This is the precise mechanism behind the co-drawdown amplification observed in Huang and Sun [2025, Figure 4].

3.5 Heterogeneous investors

We extend the analysis to a two-type game with heterogeneous risk aversion.

Proposition 3.19 (Heterogeneous investor perturbation)**.** *Consider the two-type game with N_A investors of risk aversion $\gamma_A = \gamma + \varepsilon$ and N_B investors of risk aversion $\gamma_B = \gamma - \varepsilon$, where $\varepsilon \in (-\gamma, \gamma)$ is the heterogeneity parameter. Let $\alpha_A = N_A/N$ and $\alpha_B = N_B/N$ be the population fractions. Then:*

(i) (λ-dependence at first order.) The PI strategy function $g_k(t, q; \lambda)$ for type k satisfies

$$g_k(t, q; \lambda) = g_0(t, q) + \varepsilon\, g_1^k(t, q; \lambda) + O(\varepsilon^2), \tag{30}$$

where g_0 is the symmetric (and λ-independent) function from (10), and g_1^k depends on λ through the linearized Cauchy problem

$$\partial_t c_1^k + \frac{\beta^2}{2}\,\partial_{qq} c_1^k - \frac{\beta}{\sigma}\big[\theta - \sigma^2 \bar{g}_0\big]\partial_q c_1^k + \frac{\beta\sigma\,\bar{g}_1}{}\,\partial_q c_0 + \Delta\kappa_k\,\frac{\theta^2}{\sigma^2} = 0, \tag{31}$$

with $c_1^k(T, q) = 0$, $\bar{g}_0 = g_0/[\gamma(1-\lambda)]$, $\Delta\kappa_A = -(1-\lambda/N)/\gamma^2$, and $\bar{g}_1$ is the first-order correction to the aggregate function depending on g_1^A, g_1^B, and λ.

(ii) (Wealth gap correction.) The expected wealth gap for type A satisfies

$$\frac{\partial G_A}{\partial \lambda} = \frac{G_A(\lambda, T)}{1-\lambda} + \varepsilon\, \Phi(\lambda, T) + O(\varepsilon^2), \tag{32}$$

where $\Phi(\lambda, T)$ captures the heterogeneity correction and is determined by the first-order Cauchy problem (31)*.*

(iii) (Decorrelation effect.) Under PI, heterogeneity $\varepsilon \neq 0$ implies $\pi_A^*(t) \neq \pi_B^*(t)$ at first order. The wealth correlation between types satisfies $\rho_{AB} = 1 - O(\varepsilon^2)$, so the co-drawdown probability for a mixed population is strictly less than for the symmetric case.

Remark 3.20. The key mechanism is that heterogeneity breaks the cancellation in Remark 2.3. When $\gamma_A \neq \gamma_B$, the aggregate strategy $\bar{g}$ involves a γ_k-weighted average where the more risk-tolerant type contributes disproportionately. This weighted average does not simplify to $g_0/[\gamma(1-\lambda)]$, so λ enters the Cauchy problem through the aggregate drift. The resulting λ-dependence of g_1^k is the first-order symmetry-breaking effect of heterogeneity.

Remark 3.21 (Role of the symmetry assumption). The factorization $G(\lambda, T) = \mu_1 I(T)/[\gamma(1-\lambda)]$ and the λ-independence of $g(t, q)$ both require identical (γ, λ, x) across investors. Proposition 3.19 shows that heterogeneity in γ alone introduces λ-dependence in g at first order. The exponential safety bound (Theorem 3.3) does NOT require symmetry: it holds for each individual investor.

Remark 3.22 (Role of constant drift). The binary prior $\mu \in \{\mu_1, \mu_2\}$ with constant drift is essential for the filter $P(t)$ to be a diffusion on $[0, 1]$. Under regime-switching (alternating drift), the filter becomes two-dimensional, and the hedging demand structure changes qualitatively. The core mechanism (smaller $\pi \Rightarrow$ higher $\mu/(\pi\sigma^2) \Rightarrow$ lower ruin) persists.

Remark 3.23 (Role of $r = 0$). With $r > 0$, the equilibrium strategies acquire time-dependence through $e^{-r(T-t)}$, breaking the stationarity used in the Lundberg exponent computation. The Dubins–Schwarz comparison (Theorem 3.6) extends without modification since it depends only on the quadratic variation ordering, not on stationarity.

4 Proofs

4.1 Proof of Theorem 3.3 (Exponential safety bound)

The proof constructs an exponential supermartingale with a carefully chosen tilting parameter.

Proof. Fix $T > 0$. Under $\mathbb{P}_{\mu_1}$, the PI wealth process satisfies

$$X^{\mathrm{PI}}(t) = x + \int_0^t \pi^*(s)\, \mu_1\, ds + \int_0^t \pi^*(s)\, \sigma\, dW(s), \tag{33}$$

where $\pi^*(s) = \theta(P(s))/[\sigma^2\gamma(1-\lambda)]$. Since $\theta(q) \leq \mu_1$ for $q \in [0, 1]$, we have

$$\pi^*(s) \leq \frac{\mu_1}{\sigma^2\gamma(1-\lambda)} = \pi_1^{\mathrm{FI}} =: \pi_{\max} \qquad \text{a.s., for all } s \geq 0. \tag{34}$$

Define the tilting parameter

$$\vartheta := -\frac{2\mu_1}{\pi_{\max}\,\sigma^2} \tag{35}$$

and the exponential process

$$Z(t) := \exp\!\Big(\vartheta\big(X^{\mathrm{PI}}(t) - x\big) - \int_0^t \psi(s)\,ds\Big), \tag{36}$$

where

$$\psi(s) := \vartheta\,\pi^*(s)\,\mu_1 + \frac{\vartheta^2}{2}\big(\pi^*(s)\big)^2\,\sigma^2. \tag{37}$$

Step 1: Z is a local martingale. By Itô's formula applied to $f(y) = e^{\vartheta y}$ with $y(t) = X^{\mathrm{PI}}(t) - x - \int_0^t \psi(s)/\vartheta\,ds$:

$$dZ(t) = Z(t)\,\vartheta\,\pi^*(t)\,\sigma\,dW(t). \tag{38}$$

More precisely, we verify this by writing $Z(t) = \exp(\vartheta\int_0^t \pi^*(s)[\mu_1\,ds + \sigma\,dW(s)] - \int_0^t \psi(s)\,ds)$. By the Itô exponential:

$$\begin{aligned} dZ(t) &= Z(t)\Big[\vartheta\,\pi^*(t)\,\mu_1\,dt + \vartheta\,\pi^*(t)\,\sigma\,dW(t) + \frac{\vartheta^2}{2}\,(\pi^*(t))^2\,\sigma^2\,dt - \psi(t)\,dt\Big] \\ &= Z(t)\,\vartheta\,\pi^*(t)\,\sigma\,dW(t), \end{aligned}$$

since the dt terms cancel by definition of ψ. Therefore Z is a non-negative local martingale, hence a supermartingale with $\mathbb{E}[Z(t)] \le Z(0) = 1$.

Step 2: The compensator ψ is non-positive. We show $\psi(s) \le 0$ for all s, which implies $\int_0^t \psi(s)\,ds \le 0$ and therefore $\exp(-\int_0^t \psi(s)\,ds) \ge 1$.

Factor $\psi(s)$:

$$\begin{aligned} \psi(s) &= \pi^*(s)\left[\vartheta\,\mu_1 + \frac{\vartheta^2}{2}\,\pi^*(s)\,\sigma^2\right] \\ &= \pi^*(s)\,\frac{-2\mu_1^2}{\pi_{\max}\,\sigma^2}\left[1 - \frac{\pi^*(s)}{\pi_{\max}}\right], \end{aligned} \tag{39}$$

where we used $\vartheta = -2\mu_1/(\pi_{\max}\sigma^2)$ and $\vartheta^2\sigma^2/2 = 2\mu_1^2/(\pi_{\max}^2\sigma^2)$.

Since $\pi^*(s) \ge 0$ and $\pi^*(s) \le \pi_{\max}$ by (34), the factor $[1 - \pi^*(s)/\pi_{\max}] \ge 0$, and the leading constant $-2\mu_1^2/(\pi_{\max}\sigma^2) < 0$. Therefore $\psi(s) \le 0$ for all $s \ge 0$.

Step 3: Optional stopping. Define $\tau := \tau^{\mathrm{PI}}$ for brevity. Since Z is a non-negative supermartingale:

$$1 = Z(0) \ge \mathbb{E}_{\mu_1}\big[Z(\tau \wedge T)\big]. \tag{40}$$

On the event $\{\tau \le T\}$: $X^{\mathrm{PI}}(\tau) = D$, so $X^{\mathrm{PI}}(\tau) - x = -(x - D) = -a$, and

$$Z(\tau) = \exp\!\big(-\vartheta\,a - \textstyle\int_0^\tau \psi(s)\,ds\big) \ge \exp(-\vartheta\,a) = \exp\!\Big(\frac{2\mu_1\,a}{\pi_{\max}\,\sigma^2}\Big), \tag{41}$$

where the inequality uses $\int_0^\tau \psi(s)\,ds \le 0$.

From (40):

$$\begin{aligned} 1 &\ge \mathbb{E}_{\mu_1}\big[Z(\tau)\,\mathbf{1}_{\{\tau\le T\}}\big] + \mathbb{E}_{\mu_1}\big[Z(T)\,\mathbf{1}_{\{\tau>T\}}\big] \\ &\ge \exp\!\left(\frac{2\mu_1\,a}{\pi_{\max}\,\sigma^2}\right)\cdot \mathbb{P}_{\mu_1}(\tau\le T) + 0, \end{aligned} \tag{42}$$

using $Z(T)\ge 0$ and (41). Rearranging:

$$\mathbb{P}_{\mu_1}\big(\tau^{\mathrm{PI}}\le T\big) \le \exp\!\left(-\frac{2\mu_1\,a}{\pi_{\max}\,\sigma^2}\right). \tag{43}$$

Finally, the classical first passage result for the FI process $X_1^{\mathrm{FI}}(t) = x+\pi_1^{\mathrm{FI}}\mu_1 t+\pi_1^{\mathrm{FI}}\sigma W(t)$ (Brownian motion with positive drift $b=\pi_1^{\mathrm{FI}}\mu_1$ and diffusion $c=\pi_1^{\mathrm{FI}}\sigma$) gives

$$\mathbb{P}_{\mu_1}\big(\tau_1^{\mathrm{FI}}<\infty\big) = \exp\!\left(-\frac{2b\,a}{c^2}\right) = \exp\!\left(-\frac{2\mu_1\,a}{\pi_1^{\mathrm{FI}}\,\sigma^2}\right). \tag{44}$$

Since $\pi_{\max}=\pi_1^{\mathrm{FI}}$, (43) and (44) yield (17). □

4.2 Proof of Theorem 3.6 (Martingale measure comparison)

Proof. **Step 1: Girsanov transformation.** Define Q by (18). Under Q, $\widetilde{W}(t) := W(t) + (\mu_1/\sigma)\,t$ is a standard Brownian motion. The stock has zero drift under Q: $dS/S = \sigma\,d\widetilde{W}$.

The wealth processes become:

$$X_1^{\mathrm{FI}}(t) - x = \pi_1^{\mathrm{FI}}\,\sigma\,\widetilde{W}(t), \tag{45}$$

$$X^{\mathrm{PI}}(t) - x = \sigma\int_0^t \pi^*(s)\,d\widetilde{W}(s). \tag{46}$$

Both are continuous Q-local martingales starting at 0.

Step 2: Dubins–Schwarz representation. By the Dambis–Dubins–Schwarz theorem (Karatzas and Shreve 1991, Theorem 3.4.6), there exist standard Q-Brownian motions B_1 and B_2 (possibly defined on an extension of the probability space) such that:

$$X_1^{\mathrm{FI}}(t) - x = B_1\big(U(t)\big), \qquad U(t) := \big(\pi_1^{\mathrm{FI}}\,\sigma\big)^2\,t, \tag{47}$$

$$X^{\mathrm{PI}}(t) - x = B_2\big(V(t)\big), \qquad V(t) := \sigma^2\int_0^t \big[\pi^*(s)\big]^2\,ds. \tag{48}$$

The clock U is deterministic and linear. The clock V is random and satisfies

$$V(T) = \sigma^2\int_0^T \big[\pi^*(s)\big]^2\,ds \le \sigma^2\left(\pi_1^{\mathrm{FI}}\right)^2 T = U(T), \qquad Q\text{-a.s.}, \tag{49}$$

since $\pi^*(s)\le \pi_1^{\mathrm{FI}}$ a.s. for all s. The inequality is strict with positive Q-probability whenever $\pi^*(s) < \pi_1^{\mathrm{FI}}$ on a set of positive Lebesgue measure, which holds for $p<1$.

Step 3: First passage time comparison. The ruin events translate to:

$$\{\tau_1^{\mathrm{FI}} \le T\} = \Big\{\inf_{0\le t\le T}\big(X_1^{\mathrm{FI}}(t) - x\big) \le -a\Big\} = \Big\{\inf_{0\le u\le U(T)} B_1(u) \le -a\Big\}, \tag{50}$$

$$\{\tau^{\mathrm{PI}} \le T\} = \Big\{\inf_{0\le t\le T}\big(X^{\mathrm{PI}}(t) - x\big) \le -a\Big\} = \Big\{\inf_{0\le v\le V(T)} B_2(v) \le -a\Big\}. \tag{51}$$

Since $V(T) \le U(T)$ a.s.:

$$\Big\{\inf_{0\le v\le V(T)} B_2(v) \le -a\Big\} \subseteq \Big\{\inf_{0\le v\le U(T)} B_2(v) \le -a\Big\}. \tag{52}$$

Taking Q-probabilities and using the fact that B_2 under Q has the same law as B_1 under Q (both are standard Brownian motions):

$$\begin{aligned} Q\big(\tau^{\mathrm{PI}} \le T\big) &\le Q\Big(\inf_{0\le v\le U(T)} B_2(v) \le -a\Big) \\ &= Q\Big(\inf_{0\le u\le U(T)} B_1(u) \le -a\Big) \\ &= Q\big(\tau_1^{\mathrm{FI}} \le T\big). \end{aligned} \tag{53}$$

Step 4: Equality condition. Equality in (52) holds if and only if $V(T) = U(T)$ Q-a.s., which by (49) requires $\pi^*(s) = \pi_1^{\mathrm{FI}}$ for Lebesgue-a.e. $s \in [0,T]$, Q-a.s. □

Remark 4.1. In Step 3, we used the inclusion (52) rather than a direct coupling of B_1 and B_2. This is important because the Dubins–Schwarz Brownian motions B_1 and B_2 are constructed from the respective martingales and are *not* identical processes in general. However, their marginal laws under Q are identical (both are standard Brownian motions), which suffices for the probability comparison.

Remark 4.2 (Quantifying the Girsanov gap). The gap between Theorem 3.6 (under Q) and Conjecture 3.11 (under $\mathbb{P}_{\mu_1}$) satisfies

$$\big|\mathbb{P}_{\mu_1}(\tau^I \le T) - Q(\tau^I \le T)\big| \le \sqrt{e^{m^2T} - 1}\,\sqrt{Q(\tau^I \le T)}\,, \qquad m := \mu_1/\sigma,$$

by Cauchy–Schwarz. At the calibrated parameters ($m^2T = 40$), this bound exceeds 10^8 and is thus useless. A tighter analysis decomposes $\mathbb{P}_{\mu_1}(\tau^I \le T) = \mathbb{E}_Q[L(T) \mid \tau^I \le T]\, Q(\tau^I \le T)$, where $L(T) = e^{m\widetilde{W}(T) - m^2T/2}$. The conjecture reduces to verifying $\mathbb{E}_Q[L(T) \mid \tau^{\mathrm{PI}} \le T] \le \mathbb{E}_Q[L(T) \mid \tau^{\mathrm{FI}} \le T]$: the Girsanov weight, conditional on ruin, should be smaller for PI because the PI ruin event requires a larger negative stock excursion (smaller portfolio ⇒ larger decline needed to reach the barrier), and such paths carry smaller $L(T)$.

4.3 Proof of Corollary 3.8 (Constant damping)

Proof. When $\pi^*(t) \equiv c$ for a constant $c \in (0, \pi_1^{\mathrm{FI}})$, the PI wealth process under any measure $\mathbb{P}$ equivalent to $\mathbb{P}_{\mu_1}$ satisfies

$$X^{\mathrm{PI}}(t) - x = c\big(\mu_1\, t + \sigma\, W(t)\big) = \frac{c}{\pi_1^{\mathrm{FI}}}\,\big(X_1^{\mathrm{FI}}(t) - x\big). \tag{54}$$

Define $\alpha := c/\pi_1^{\mathrm{FI}} \in (0,1)$. Then $X^{\mathrm{PI}}(t) = (1-\alpha)\,x + \alpha\, X_1^{\mathrm{FI}}(t)$ and

$$X^{\mathrm{PI}}(t) \le D \iff X_1^{\mathrm{FI}}(t) \le \frac{D-(1-\alpha)\,x}{\alpha} = x - \frac{a}{\alpha}\,. \tag{55}$$

Since $\alpha < 1$, the effective barrier $x - a/\alpha < x - a = D$ is strictly below D. Therefore

$$\big\{\tau^{\mathrm{PI}} \le T\big\} = \Big\{\inf_{t\le T} X_1^{\mathrm{FI}}(t) \le x - \frac{a}{\alpha}\Big\} \subsetneq \big\{\tau_1^{\mathrm{FI}} \le T\big\}, \tag{56}$$

and the inclusion is strict because the event $\{x - a/\alpha < \inf_{t\le T} X_1^{\mathrm{FI}}(t) \le D\}$ has positive probability. Taking probabilities under any measure gives (20). □

4.4 Proof of Proposition 3.13

We prove that under $\mathbb{P}_{\mu_2}$ (the bad state), the PI investor over-invests relative to the FI optimal: $\mathbb{E}_{\mu_2}[\pi^*(t)] > \pi_2^{\mathrm{FI}}$ for all $t > 0$ and $p \in (0,1)$.

Proof. The proof proceeds in four steps.

Step 1: Filter dynamics under $\mathbb{P}_{\mu_2}$. Under $\mathbb{P}_{\mu_2}$, the stock return is $dR(t) = \mu_2\,dt + \sigma\,dW(t)$, where W is a standard $\mathbb{P}_{\mu_2}$-Brownian motion. The Bayesian filter (7) admits the Bayes formula representation

$$P(t) = \frac{1}{1 + \frac{1-p}{p}\,\Lambda(t)}, \tag{57}$$

where $\Lambda(t) := \ell_2(t)/\ell_1(t)$ is the likelihood ratio with

$$\Lambda(t) = \exp\!\Big(-\frac{\delta\,W(t)}{\sigma} + \frac{\delta^2\,t}{2\sigma^2}\Big), \qquad \delta := \mu_1 - \mu_2. \tag{58}$$

Indeed, the likelihood functions satisfy $\ell_k(t) = \exp\big(\mu_k R(t)/\sigma^2 - \mu_k^2 t/(2\sigma^2)\big)$. Substituting $R(t) = \mu_2 t + \sigma W(t)$ and simplifying:

$$\frac{\ell_2(t)}{\ell_1(t)} = \exp\!\Big(\frac{(\mu_2-\mu_1)R(t)}{\sigma^2} + \frac{\mu_1^2-\mu_2^2}{2\sigma^2}\,t\Big) = \exp\!\Big(-\frac{\delta\,W(t)}{\sigma} + \frac{\delta^2\,t}{2\sigma^2}\Big),$$

using $(\mu_1+\mu_2)/2 - \mu_2 = \delta/2$.

Step 2: $P(t) > 0$ almost surely for finite t. Since $W(t)$ is finite a.s. for each fixed t, we have $\Lambda(t) < \infty$ a.s., hence $P(t) = 1/(1 + \frac{1-p}{p}\Lambda(t)) > 0$ a.s. for all $t \in [0,\infty)$. As $t \to \infty$, the strong law of large numbers gives $W(t)/t \to 0$ a.s., so $\Lambda(t) \to +\infty$ a.s. (the drift $\delta^2 t/(2\sigma^2)$ dominates), yielding $P(t) \to 0$ a.s. However, this convergence requires infinite time; for any finite t, $P(t) > 0$ a.s.

Step 3: The filter is a $\mathbb{P}_{\mu_2}$-supermartingale with $\mathbb{E}_{\mu_2}[P(t)] > 0$. Under the mixture measure $\mathbb{P}_p := p\,\mathbb{P}_{\mu_1} + (1-p)\,\mathbb{P}_{\mu_2}$, the filter $P(t) = \mathbb{P}_p(\mu = \mu_1 \mid \mathcal{F}_t^S)$ is a $\mathbb{P}_p$-martingale by the tower property:

$$p = \mathbb{E}_{\mathbb{P}_p}[P(t)] = p\,\mathbb{E}_{\mu_1}[P(t)] + (1-p)\,\mathbb{E}_{\mu_2}[P(t)]. \tag{59}$$

Under $\mathbb{P}_{\mu_1}$, the filter learns the truth: $P(t) \to 1$ a.s. as $t \to \infty$, so $\mathbb{E}_{\mu_1}[P(t)] > p$ for $t > 0$ (the filter is a $\mathbb{P}_{\mu_1}$-submartingale, as established in the Bayes formula (57)). Substituting into (59): $(1-p)\,\mathbb{E}_{\mu_2}[P(t)] < p - p \cdot p = p(1-p)$, which gives $\mathbb{E}_{\mu_2}[P(t)] < p$. Together with the pointwise bound $P(t) > 0$ a.s., we conclude $\mathbb{E}_{\mu_2}[P(t)] \in (0, p)$ for all $t > 0$.

Step 4: Over-investment. Since $\theta(q) = \delta q + \mu_2$ is affine:

$$\mathbb{E}_{\mu_2}[\theta(P(t))] = \delta\,\mathbb{E}_{\mu_2}[P(t)] + \mu_2. \tag{60}$$

By Step 3, $\mathbb{E}_{\mu_2}[P(t)] > 0$, hence

$$\mathbb{E}_{\mu_2}[\pi^*(t)] = \frac{\mathbb{E}_{\mu_2}[\theta(P(t))]}{\sigma^2\gamma(1-\lambda)} = \frac{\delta\,\mathbb{E}_{\mu_2}[P(t)] + \mu_2}{\sigma^2\gamma(1-\lambda)} > \frac{\mu_2}{\sigma^2\gamma(1-\lambda)} = \pi_2^{\mathrm{FI}}$$

for all $t > 0$ and $p \in (0,1)$. □

Remark 4.3 (Initial over-investment ratio). At $t = 0$: $\pi^*(0)/\pi_2^{\mathrm{FI}} = (\delta p + \mu_2)/\mu_2 = 1 + \delta p/\mu_2$. For the Huang–Sun parameters ($\delta = 0.18, p = 0.5, \mu_2 = 0.02$), this ratio equals 5.5. The PI investor initially invests more than five times the FI optimal under the bad state, because her prior assigns probability $p = 0.5$ to the good state $\mu_1 = 0.2$ while the true drift is only $\mu_2 = 0.02$.

Lemma 4.4 (Filter modulus of continuity)**.** *Under either $\mathbb{P}_{\mu_k}$, the filter satisfies*

$$\mathbb{E}\big[|P(s) - P(t)|^2 \,\big|\, \mathcal{F}_t\big] \;\le\; \frac{\delta^2}{16\sigma^2}\,(s-t), \qquad 0 \le t \le s,$$

and consequently $\mathbb{E}\big[\sup_{s\in[t,t+h]} |P(s) - P(t)|^2\big] \le \delta^2 h/(4\sigma^2)$.

Proof. By Itô isometry, $\mathbb{E}[|P(s) - P(t)|^2 \mid \mathcal{F}_t] = \mathbb{E}[\int_t^s \beta(P(u))^2\,du \mid \mathcal{F}_t]$ where $\beta(q) = (\delta/\sigma)q(1-q)$. Since $\max_{q\in[0,1]} q^2(1-q)^2 = 1/16$, we have $\beta(q)^2 \le \delta^2/(16\sigma^2)$. The supremum bound follows from Doob's L^2-maximal inequality. □

Remark 4.5. Lemma 4.4 applies to any binary-state Bayesian filtering problem with Gaussian observations. The bound $\delta^2/(16\sigma^2)$ depends only on the signal-to-noise ratio δ/σ and is independent of the true state, prior, or time. This makes it a reusable tool for L^2-coupling arguments involving Bayesian filters.

4.5 Proof of Theorem 3.14

We prove the bad-state ruin comparison in three stages: (A) exact FI ruin probability, (B) PI ruin lower bound via L^2 coupling with a constant-portfolio reference, and (C) amplification ratio computation.

Proof. **Stage A: Exact FI ruin probability.**

Under $\mathbb{P}_{\mu_2}$, the FI wealth deviation $Y_2^{\mathrm{FI}}(t) := X_2^{\mathrm{FI}}(t) - x = \pi_2^{\mathrm{FI}}(\mu_2 t + \sigma W(t))$ is Brownian motion with drift $b := \pi_2^{\mathrm{FI}}\mu_2 > 0$ and diffusion $c := \pi_2^{\mathrm{FI}}\sigma > 0$. By the classical first-passage

formula [Karatzas and Shreve, 1991, Proposition 3.5.2], the infinite-horizon ruin probability is

$$\mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty) = \exp\!\left(-\frac{2b\,a}{c^2}\right) = \exp\!\left(-\frac{2\mu_2\,a}{\pi_2^{\mathrm{FI}}\sigma^2}\right) = \exp\!\big(-2\gamma(1-\lambda)\,a\big), \tag{61}$$

where $a := x - D > 0$ is the safety margin and we used $\pi_2^{\mathrm{FI}} = \mu_2/[\sigma^2\gamma(1-\lambda)]$.

Stage B: PI ruin lower bound via coupling.

The argument compares the PI wealth process to a constant-portfolio reference process over a short time window, using L^2 proximity. We proceed through four substeps.

Step B1: Reference process and BL monotonicity. Define the constant-portfolio reference wealth:

$$\bar{\pi} := \pi^*(0) = \frac{\theta(p)}{\sigma^2\gamma(1-\lambda)} = \frac{\delta p + \mu_2}{\sigma^2\gamma(1-\lambda)}. \tag{62}$$

Note that $\bar{\pi}/\pi_2^{\mathrm{FI}} = (\delta p + \mu_2)/\mu_2 = 1 + \delta p/\mu_2 > 1$. The reference process is $\widetilde{Z}(t) := \bar{\pi}(\mu_2 t + \sigma W(t))$, which is Brownian motion with drift $\bar{\pi}\mu_2$ and diffusion $\bar{\pi}\sigma$.

Lemma 4.6 (Bachelier–Lévy monotonicity)**.** *For a constant portfolio $\pi > 0$ and the process $Z_\pi(t) := \pi(\mu_2 t + \sigma B(t))$, the ruin probability*

$$\mathfrak{p}(a, \pi, T) := \mathbb{P}\big(\inf_{t \le T} Z_\pi(t) < -a\big)$$

is strictly increasing in π for each $a > 0$ and $T > 0$.

Proof of Lemma 4.6. Write $\inf_{t\le T} Z_\pi(t) = \pi \inf_{t\le T}(\mu_2 t + \sigma B(t)) =: \pi\, m_T$, where $m_T < 0$ a.s. (since Brownian motion takes negative values near time 0). Then $\{Z_\pi < -a\} = \{\pi\, m_T < -a\} = \{m_T < -a/\pi\}$. Since a/π is strictly decreasing in π and $\mathbb{P}(m_T < -c)$ is strictly decreasing in $c > 0$ (as m_T has a continuous distribution), $\mathfrak{p}(a, \pi, T)$ is strictly increasing in π. □

By Lemma 4.6, since $\bar{\pi} > \pi_2^{\mathrm{FI}}$:

$$\mathbb{P}_{\mu_2}(\widetilde{\tau} \le T) = \mathfrak{p}(a, \bar{\pi}, T) > \mathfrak{p}(a, \pi_2^{\mathrm{FI}}, T) = \mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} \le T) \tag{63}$$

for all $T > 0$, where $\widetilde{\tau} := \inf\{t \ge 0 : \widetilde{Z}(t) < -a\}$. In the limit $T \to \infty$:

$$R(\bar{\pi}) := \mathbb{P}_{\mu_2}(\widetilde{\tau} < \infty) = \exp\!\left(-\frac{2\mu_2\,a}{\bar{\pi}\sigma^2}\right) > \exp\!\left(-\frac{2\mu_2\,a}{\pi_2^{\mathrm{FI}}\sigma^2}\right) = \mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty). \tag{64}$$

Step B2: Filter modulus of continuity. Under $\mathbb{P}_{\mu_2}$, the filter satisfies the SDE

$$dP(t) = \frac{\delta\, P(t)(1-P(t))}{\sigma}\, d\widehat{W}(t), \qquad P(0) = p, \tag{65}$$

where $\widehat{W}$ is the innovation process (a standard Brownian motion under $\mathbb{P}_{\mu_2}$). Define $\sigma_P := \delta\, p(1-p)/\sigma$ (the initial filter volatility). For short times $s \le 1$:

$$\mathbb{E}_{\mu_2}\big[|P(s) - p|^2\big] = \sigma_P^2\, s + O(s^2) \le C_P\, s, \tag{66}$$

where $C_P := 2\delta^2 p^2(1-p)^2/\sigma^2$.

Since $\pi^*(t) = (\delta\, P(t) + \mu_2)/[\sigma^2\gamma(1-\lambda)]$ (myopic approximation), the strategy modulus satisfies

$$\mathbb{E}_{\mu_2}\big[|\pi^*(s) - \bar{\pi}|^2\big] = \frac{\delta^2}{\sigma^4\gamma^2(1-\lambda)^2}\,\mathbb{E}_{\mu_2}\big[|P(s) - p|^2\big] \le C_\pi\, s, \tag{67}$$

where $C_\pi := 2\delta^4 p^2(1-p)^2/[\sigma^6\gamma^2(1-\lambda)^2]$.

Step B3: L^2 bound on the process difference. Let $Z^{\mathrm{PI}}(t) := X^{\mathrm{PI}}(t) - x$ be the PI wealth deviation and define $\Delta(t) := Z^{\mathrm{PI}}(t) - \widetilde{Z}(t)$. Then

$$\Delta(t) = \int_0^t (\pi^*(s) - \bar{\pi})\big(\mu_2\, ds + \sigma\, dW(s)\big)$$

is a semimartingale. By Doob's L^2-maximal inequality applied to the martingale part $M_\Delta(t) := \int_0^t(\pi^*(s) - \bar{\pi})\sigma\, dW(s)$ together with the Itô isometry, and by Cauchy–Schwarz for the drift part $A_\Delta(t) := \int_0^t(\pi^*(s) - \bar{\pi})\mu_2\, ds$:

$$\mathbb{E}_{\mu_2}\Big[\sup_{s\le h}|\Delta(s)|^2\Big] \le 4\sigma^2 C_\pi\, h^2 + \mu_2^2 C_\pi\, h^3 \le K\, h^2 \tag{68}$$

for $h \le 1$, where $K := C_\pi(4\sigma^2 + \mu_2^2)$.

By Markov's inequality:

$$\mathbb{P}_{\mu_2}\Big(\sup_{s\le h}|\Delta(s)| > \eta\Big) \le \frac{K\, h^2}{\eta^2}. \tag{69}$$

Step B4: Ruin probability approximation. For any $\eta > 0$ and $h > 0$, we have the event inclusion

$$\big\{\inf_{s\le h}\widetilde{Z}(s) < -(a+\eta)\big\} \cap \big\{\sup_{s\le h}|\Delta(s)| \le \eta\big\} \;\subseteq\; \big\{\inf_{s\le h} Z^{\mathrm{PI}}(s) < -a\big\}.$$

Indeed, on this event: $\inf Z^{\mathrm{PI}} \le \inf\widetilde{Z} + \sup|\Delta| < -(a+\eta) + \eta = -a$. Taking probabilities and using $\mathbb{P}(A\cap B) \ge \mathbb{P}(A) - \mathbb{P}(B^c)$:

$$\mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} \le h) \ge \mathfrak{p}\big((1+\epsilon)a,\ \bar{\pi}\mu_2,\ \bar{\pi}\sigma,\ h\big) - \frac{K\, h^2}{\epsilon^2 a^2}, \tag{70}$$

where we set $\eta = \epsilon\, a$ for a parameter $\epsilon \in (0, \delta p/\mu_2)$.

Stage C: Amplification ratio.

We now show that the right-hand side of (70) exceeds $\mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty)$ for appropriate h.

By the Bachelier–Lévy formula [Karatzas and Shreve, 1991, Proposition 3.5.2]:

$$\mathfrak{p}(a', b, c, T) = \Phi\Big(\frac{-a' + bT}{c\sqrt{T}}\Big) + e^{-2a'b/c^2}\,\Phi\Big(\frac{-a' - bT}{c\sqrt{T}}\Big), \tag{71}$$

where Φ is the standard normal CDF. As $T \to \infty$: the first Φ-term converges to 1 and the second to 0, so $\mathfrak{p}(a', b, c, T) \to e^{-2a'b/c^2}$.

With $a' = (1+\epsilon)a$, $b = \bar{\pi}\mu_2$, $c = \bar{\pi}\sigma$:

$$R_\epsilon := \lim_{T\to\infty} \mathfrak{p}\big((1+\epsilon)a,\ \bar{\pi}\mu_2,\ \bar{\pi}\sigma,\ T\big) = \exp\!\left(-\frac{2(1+\epsilon)\mu_2\, a}{\bar{\pi}\sigma^2}\right). \tag{72}$$

We need $R_\epsilon > R(\pi_2^{\mathrm{FI}})$, i.e., $(1+\epsilon)/\bar{\pi} < 1/\pi_2^{\mathrm{FI}}$, i.e., $\epsilon < \bar{\pi}/\pi_2^{\mathrm{FI}} - 1 = \delta p/\mu_2$. This holds by hypothesis. Define $\Delta R := R_\epsilon - R(\pi_2^{\mathrm{FI}}) > 0$.

Choosing h: Since $\mathfrak{p}$ is continuous and converges to R_ϵ as $T \to \infty$, there exists $h_1 > 0$ such that for all $h \geq h_1$:

$$\mathfrak{p}\big((1+\epsilon)a,\ \bar{\pi}\mu_2,\ \bar{\pi}\sigma,\ h\big) \geq R_\epsilon - \Delta R/4.$$

Choose $h_2 > 0$ such that $Kh_2^2/(\epsilon^2 a^2) = \Delta R/4$, i.e., $h_2 = \epsilon\, a\, \sqrt{\Delta R/(4K)}$. Setting $h = \max(h_1, h_2)$ in (70):

$$\begin{aligned}
\mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} \leq h) &\geq \big(R_\epsilon - \Delta R/4\big) - \Delta R/4 \\
&= R_\epsilon - \Delta R/2 \\
&> R(\pi_2^{\mathrm{FI}}) = \mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty).
\end{aligned}$$

Since $\mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} < \infty) \geq \mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} \leq h)$, this establishes part (i) of the theorem.

Amplification ratio. Taking $\epsilon \to 0$ in the ratio of ruin probabilities:

$$\frac{\mathbb{P}_{\mu_2}(\tau^{\mathrm{PI}} < \infty)}{\mathbb{P}_{\mu_2}(\tau_2^{\mathrm{FI}} < \infty)} \geq C \cdot \frac{R(\bar{\pi})}{R(\pi_2^{\mathrm{FI}})} = C \cdot \exp\!\left(\frac{2\mu_2\, a}{\sigma^2}\left(\frac{1}{\pi_2^{\mathrm{FI}}} - \frac{1}{\bar{\pi}}\right)\right), \tag{73}$$

where $C \in (0,1)$ accounts for the L^2 error terms. The exponent simplifies:

$$\begin{aligned}
\frac{2\mu_2\, a}{\sigma^2}\left(\frac{1}{\pi_2^{\mathrm{FI}}} - \frac{1}{\bar{\pi}}\right) &= \frac{2\mu_2\, a}{\sigma^2} \cdot \sigma^2\gamma(1-\lambda) \cdot \frac{\delta p}{\mu_2(\delta p + \mu_2)} \\
&= 2\gamma(1-\lambda)\, a\, \frac{\delta p}{\delta p + \mu_2}.
\end{aligned} \tag{74}$$

This completes the proof. □

Remark 4.7 (Numerical illustration of the amplification ratio). For the Huang–Sun parameters ($\mu_1 = 0.2, \mu_2 = 0.02, \sigma = 0.1, \gamma = 5, \lambda = 0.3, p = 0.5, a = 1$):

- $\bar{\pi}/\pi_2^{\mathrm{FI}} = 5.5$;
- $R(\pi_2^{\mathrm{FI}}) = e^{-7.00} \approx 9.1 \times 10^{-4}$;
- $R(\bar{\pi}) = e^{-1.27} \approx 0.280$;
- Exponential amplification: $R(\bar{\pi})/R(\pi_2^{\mathrm{FI}}) = e^{5.73} \approx 307$.

Even with a conservative constant $C = 0.1$ in (73), the PI ruin probability exceeds the FI one by a factor of at least 30.

Remark 4.8 (Comparison with finite-horizon numerics). The infinite-horizon bound in Theorem 3.14 yields an amplification ratio of ~ 307 at $a = 1$, while the Monte Carlo simulations at finite horizon $T = 10$ (Table 2) show ratios of 2–3. The discrepancy arises because the finite-horizon ruin probability is substantially smaller than the infinite-horizon one, and the coupling error term $Kh^2/(\epsilon^2 a^2)$ is non-negligible for moderate h. The theorem correctly predicts the *direction* (PI riskier under the bad state) and the *mechanism* (over-investment), while the precise finite-horizon ratio requires numerical computation.

4.6 Proof of Theorem 3.17

We prove that the PI ruin probability is more sensitive to the competition parameter λ than the FI ruin probability, with an explicit amplification factor.

Proof. The key structural fact, established in Remark 2.3, is that the function $g(t, q)$ does not depend on λ under the symmetric game. Consequently, the PI strategy is $\pi^*(t) = g(t, P(t))/[\gamma(1-\lambda)]$, and the dependence on λ enters *only* through the explicit prefactor $1/[\gamma(1-\lambda)]$.

Step 1: FI sensitivity. Under $\mathbb{P}_{\mu_1}$ (the good state), the FI wealth is a drifted Brownian motion with $\pi^{\mathrm{FI}} = \mu_1/[\sigma^2\gamma(1-\lambda)]$. The infinite-horizon ruin probability is

$$\mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} < \infty) = \exp\left(-\frac{2\mu_1\, a}{\pi^{\mathrm{FI}}\sigma^2}\right) = e^{-2\gamma(1-\lambda)a}.$$

Define $\psi^{\mathrm{FI}} := 2\gamma(1-\lambda)$, so that $-\log \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} < \infty) = \psi^{\mathrm{FI}} \cdot a$. The sensitivity is

$$\frac{d}{d\lambda}\big(-\log \mathbb{P}(\tau^{\mathrm{FI}} < \infty)\big) = -2\gamma\, a. \tag{75}$$

This is negative: increasing λ increases FI ruin probability, as expected from the portfolio amplification $\pi^{\mathrm{FI}} \propto 1/(1-\lambda)$.

Step 2: PI Lundberg exponent. For the PI investor, the effective "Lundberg exponent" governing the exponential decay of ruin probability with the safety margin a is

$$\psi^{\mathrm{PI}} := -\frac{\log \mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} < \infty)}{a}. \tag{76}$$

By Theorem 3.3, $\psi^{\mathrm{PI}} \geq \psi^{\mathrm{FI}}$ (PI is exponentially safer under the good state).

The exponential martingale bound (Theorem 3.3) gives

$$\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \leq T) \leq \exp\left(-\frac{2\mu_1\, a}{\pi_{\max}\, \sigma^2}\right),$$

where $\pi_{\max} = \mu_1/[\sigma^2\gamma(1-\lambda)]$ (the maximum possible PI portfolio, attained when $P(t) = 1$). Since $\pi_{\max} = \pi^{\mathrm{FI}}$, the bound recovers the FI infinite-horizon probability.

For the myopic approximation (ignoring the hedging term), the initial PI portfolio is $\bar{\pi} = \theta(p)/[\sigma^2\gamma(1-\lambda)]$. The effective Lundberg exponent for the PI investor under the myopic approximation satisfies

$$\psi^{\mathrm{PI}}_{\mathrm{myopic}} \approx \frac{2\mu_1}{\bar{\pi}_0\, \sigma^2} = \frac{2\mu_1\gamma(1-\lambda)}{\theta(p)}, \tag{77}$$

where $\bar{\pi}_0 := \theta(p)/[\sigma^2\gamma(1-\lambda)]$ is the initial PI portfolio. Since $\theta(p) = \delta p + \mu_2 < \mu_1$ for $p < 1$, we have $\psi^{\mathrm{PI}}_{\mathrm{myopic}} > \psi^{\mathrm{FI}}$.

Step 3: Sensitivity ratio. Both Lundberg exponents are proportional to $(1-\lambda)$:

$$\psi^{\mathrm{FI}} = 2\gamma(1-\lambda), \qquad \psi^{\mathrm{PI}}_{\mathrm{myopic}} = \frac{2\mu_1\gamma(1-\lambda)}{\theta(p)}.$$

The ratio is

$$\frac{\psi^{\mathrm{PI}}_{\mathrm{myopic}}}{\psi^{\mathrm{FI}}} = \frac{\mu_1}{\theta(p)} = \frac{\mu_1}{\delta p + \mu_2}. \tag{78}$$

For the Huang–Sun parameters: $\mu_1/\theta(0.5) = 0.2/0.11 \approx 1.82$.

The *sensitivity* of $-\log \mathbb{P}(\tau < \infty)$ to λ is proportional to ψ, so the PI sensitivity exceeds FI by the same factor:

$$\frac{d(-\log \mathbb{P}(\tau^{\mathrm{PI}}))/d\lambda}{d(-\log \mathbb{P}(\tau^{\mathrm{FI}}))/d\lambda} = \frac{\psi^{\mathrm{PI}}}{\psi^{\mathrm{FI}}} > 1. \tag{79}$$

Step 4: Probability-level amplification. In terms of the ruin probabilities themselves, for small changes in λ:

$$\begin{aligned} \frac{d\log \mathbb{P}(\tau^{\mathrm{PI}})}{d\lambda} &= \psi^{\mathrm{PI}} \cdot a \cdot \frac{1}{1-\lambda} \qquad \text{(approximately)}, \\ \frac{d\log \mathbb{P}(\tau^{\mathrm{FI}})}{d\lambda} &= 2\gamma\, a. \end{aligned} \tag{80}$$

The ratio of the λ-elasticities of the ruin probabilities equals the Lundberg exponent ratio (78). □

Remark 4.9 (Numerical verification of the sensitivity ratio). Our Monte Carlo simulations (Table 4) yield empirical Lundberg exponents $\psi^{\mathrm{PI}}_{\mathrm{eff}} \approx 30$–$66$ across different λ values, compared with theoretical $\psi^{\mathrm{FI}} = 2\gamma(1-\lambda) \in [5.5, 10]$. The empirical ratio $\psi^{\mathrm{PI}}/\psi^{\mathrm{FI}} \approx 5$–$6$ exceeds the myopic prediction of 1.82, consistent with the additional safety provided by the hedging term, which further reduces the effective PI portfolio.

4.7 Proof of Proposition 3.19

We outline the perturbation analysis for heterogeneous risk aversion. Full algebraic details are provided in the supplementary appendix.

Proof sketch. Consider two types of investors: type A with risk aversion $\gamma_A = \gamma + \varepsilon$ and type B with $\gamma_B = \gamma - \varepsilon$, each comprising $N/2$ of the N investors. At $\varepsilon = 0$, this reduces to the symmetric game.

Step 1: Coupled PDE system. Under heterogeneity, the equilibrium functions g_A and g_B satisfy a coupled Cauchy system [Huang and Sun, 2025, Theorems 3.1 and 3.2, Remark 3.4]:

$$\begin{cases} \partial_t c_k + \dfrac{\beta^2}{2}\,\partial_{qq} c_k - \dfrac{\beta}{\sigma}\left[\theta - \sigma^2 \bar{g}(t,q)\right]\partial_q c_k + \kappa_k \dfrac{\theta^2}{\sigma^2} = 0, \\ c_k(T,q) = 0, \qquad k \in \{A, B\}, \end{cases} \tag{81}$$

where the aggregate function $\bar{g} = \frac{1}{N(1-\lambda)}\left[\frac{N_A g_A}{\gamma_A} + \frac{N_B g_B}{\gamma_B}\right]$ couples the two equations through the drift coefficient.

Step 2: Perturbation expansion. Write $c_k = c_0 + \varepsilon\, c_1^k + O(\varepsilon^2)$ and $g_k = g_0 + \varepsilon\, g_1^k + O(\varepsilon^2)$. At zeroth order, c_0 and g_0 satisfy the symmetric Cauchy problem and are λ-independent (Remark 2.3).

The first-order corrections satisfy

$$g_1^A = -\frac{\beta}{\sigma\kappa_0}\,\partial_q c_1^A - \frac{\beta\,\partial_q c_0}{\sigma\kappa_0\gamma}, \tag{82}$$

$$g_1^B = -\frac{\beta}{\sigma\kappa_0}\,\partial_q c_1^B + \frac{\beta\,\partial_q c_0}{\sigma\kappa_0\gamma}, \tag{83}$$

where $\kappa_0 = 1/\gamma$.

Step 3: Antisymmetry and decoupling. With $N_A = N_B = N/2$, the aggregate first-order correction vanishes: $\bar{g}_1 = (g_1^A + g_1^B)/[2\gamma(1-\lambda)] = 0$, because $g_1^A = -g_1^B$ (from the antisymmetry $c_1^A = -c_1^B$). The antisymmetric mode $c_1^D := c_1^A$ satisfies the *decoupled* PDE

$$\mathcal{L}^0 c_1^D + \frac{(1-\lambda/N)\,\theta^2}{\gamma^2\sigma^2} = 0, \qquad c_1^D(T,q) = 0, \tag{84}$$

where $\mathcal{L}^0$ is the backward Kolmogorov operator of the zeroth-order filter process.

Step 4: Stochastic representation and λ-dependence. By the Feynman–Kac theorem, the solution of (84) has the representation

$$c_1^D(t,q) = \frac{(1-\lambda/N)}{\gamma^2\sigma^2}\,\mathbb{E}_{Q_0}^{t,q}\!\left[\int_t^T \theta\big(P_0(u)\big)^2\,du\right] > 0, \tag{85}$$

where P_0 is the filter process under the zeroth-order Girsanov measure Q_0. The source term $\theta^2 > 0$ ensures strict positivity.

The λ-dependence enters through:

(a) the explicit prefactor $(1-\lambda/N)$, and

(b) the measure Q_0, whose drift coefficient involves $\bar{g}_0 = g_0/[\gamma(1-\lambda)]$.

Both channels are absent in the symmetric case, confirming that heterogeneity breaks the λ-independence of g.

Step 5: Effect on the wealth gap. The expected wealth gap for type A at first order in ε is

$$G_A(\lambda,T) = G_0(\lambda,T) - \frac{\varepsilon\mu_1}{\gamma(1-\lambda)}\left[\frac{I_0^{\mathrm{tot}}}{\gamma} + I_1^{A,\mathrm{tot}}\right] + O(\varepsilon^2), \tag{86}$$

where $I_0^{\mathrm{tot}} = \int_0^T [\mu_1/\sigma^2 - \mathbb{E}_{\mu_1}[g_0(t,P(t))]]\,dt > 0$ and $I_1^{A,\mathrm{tot}} = \int_0^T \mathbb{E}_{\mu_1}[g_1^A(t,P(t))]\,dt$.

The λ-derivative satisfies

$$\frac{\partial G_A}{\partial\lambda} = \frac{G_A}{1-\lambda} + \varepsilon\,\Phi(\lambda,T) + O(\varepsilon^2), \tag{87}$$

where $\Phi(\lambda,T)$ captures the λ-correction from heterogeneity. The sign of Φ is determined by the stochastic representation of c_1^D: since the source term in (84) is proportional to $(1-\lambda/N)$ and the measure Q_0 makes $\mathbb{E}_{Q_0}[\theta^2]$ increasing in λ (through the drift modification), we obtain $\Phi(\lambda,T) > 0$ in the mean-field limit $N \to \infty$. □

5 Numerical Analysis

We validate the analytical results of Section 3 through extensive Monte Carlo simulations. All experiments use the parameter calibration of Huang and Sun [2025, Figure 4], and we simulate both FI and PI strategies on the *same* Brownian paths to minimize variance in comparisons.

5.1 Simulation setup

Base parameters. $\mu_1 = 0.2$, $\mu_2 = 0.02$, $\sigma = 0.1$, $\gamma = 5$, $x = 3$, $p = 0.5$, $T = 10$. The competition parameter λ varies from 0 to 0.5. The warning threshold D is varied from 1.5 to 2.95.

Monte Carlo design. We simulate $M = 200{,}000$ paths (increased to 500,000 for the Lundberg exponent analysis) with time step $\Delta t = 0.01$ using the Euler–Maruyama scheme. Both FI and PI strategies are simulated using the *same* Brownian increments $\{\Delta W_k\}_{k=1}^{T/\Delta t}$, so that the difference $X^{\mathrm{FI}}(t) - X^{\mathrm{PI}}(t)$ has reduced variance.

PI strategy approximation. We use the *myopic* PI strategy $\pi^*(t) = \theta(P(t))/[\sigma^2\gamma(1-\lambda)]$, omitting the hedging term $h(t,q) \geq 0$. By the hedging demand analysis in (10), this overestimates the PI investment relative to the true equilibrium strategy. For the good-state comparison ($\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \leq T) \leq \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} \leq T)$), the myopic approximation is *conservative*: it makes the PI investor more aggressive than the true equilibrium, bringing PI closer to FI. Any safety advantage found for the myopic PI thus also holds for the true PI. For the bad-state comparison, the myopic approximation slightly underestimates the over-investment effect (since the hedging term further reduces π^* below the myopic level).

Simulation under $\mathbb{P}_{\mu_k}$. To condition on the true state, we set the drift in the stock SDE to $\mu = \mu_k$ for $k \in \{1, 2\}$. The filter $P(t)$ is updated using the Euler–Maruyama discretization of the innovation process, with the true state determining the relationship between the stock return and the innovation.

5.2 Good-state safety

Table 1 presents the ruin probabilities $\mathbb{P}_{\mu_1}(\tau < T)$ for FI and PI investors under the good state.

Table 1: Ruin probabilities under $\mathbb{P}_{\mu_1}$ (good state). Parameters: $\mu_1 = 0.2$, $\mu_2 = 0.02$, $\sigma = 0.1$, $\gamma = 5$, $x = 3$, $p = 0.5$, $T = 10$. Monte Carlo with $M = 200{,}000$ paths.

D	λ	$\mathbb{P}^{\mathrm{FI}}(\tau < T)$	$\mathbb{P}^{\mathrm{PI}}(\tau < T)$	Ratio
2.0	0.0	0.000040	$<5\times10^{-6}$	≈ 0
2.0	0.1	0.000115	$<5\times10^{-6}$	≈ 0
2.0	0.2	0.000320	$<5\times10^{-6}$	≈ 0
2.0	0.3	0.000795	$<5\times10^{-6}$	≈ 0
2.5	0.0	0.005360	$<5\times10^{-6}$	≈ 0
2.5	0.1	0.008870	$<5\times10^{-6}$	≈ 0
2.5	0.2	0.014700	$<5\times10^{-6}$	≈ 0
2.5	0.3	0.024050	$<5\times10^{-6}$	≈ 0
2.8	0.0	0.107305	$<5\times10^{-6}$	≈ 0
2.8	0.1	0.130590	0.000010	< 0.001
2.8	0.2	0.160225	0.000055	< 0.001
2.8	0.3	0.195515	0.000230	0.001
2.9	0.0	0.291390	0.006685	0.023
2.9	0.1	0.322010	0.014015	0.044
2.9	0.2	0.355450	0.027360	0.077
2.9	0.3	0.393355	0.049635	0.126

Discussion. The PI investor is dramatically safer than the FI investor under the good state, with zero observed ruin for $D \leq 2.8$ across all λ values in 200,000 simulated paths. Even at $D = 2.9$ (safety margin $a = 0.1$), the PI ruin probability is 2%–13% of the FI level depending on λ. This is consistent with Theorem 3.3: the PI ruin probability is bounded by the FI *infinite-horizon* ruin probability, which for $a \geq 0.2$ is vanishingly small.

The mechanism is transparent: the PI investor starts at $\pi^*(0) = 2.20$ (at $\lambda = 0$) compared to $\pi^{\mathrm{FI}} = 4.00$, giving 45% less leverage. This reduced leverage during the early learning phase—when the filter is still uncertain—provides an exponential safety buffer.

The ratio $\mathbb{P}^{\mathrm{PI}}/\mathbb{P}^{\mathrm{FI}}$ increases with λ (from 0.023 to 0.126 at $D = 2.9$), indicating that competition erodes PI's safety advantage, but never eliminates it within the parameter range studied.

5.3 Bad-state amplification

Table 2 presents the ruin probabilities under the bad state $\mathbb{P}_{\mu_2}$.

Table 2: Ruin probabilities under $\mathbb{P}_{\mu_2}$ (bad state). Same parameters as Table 1.

D	λ	$\mathbb{P}^{\mathrm{FI}}(\tau < T)$	$\mathbb{P}^{\mathrm{PI}}(\tau < T)$	Ratio
2.0	0.0	$<5\times10^{-6}$	$<5\times10^{-6}$	—
2.0	0.1	$<5\times10^{-6}$	$<5\times10^{-6}$	—
2.0	0.2	$<5\times10^{-6}$	$<5\times10^{-6}$	—
2.0	0.3	$<5\times10^{-6}$	$<5\times10^{-6}$	—
2.5	0.0	0.000010	0.000010	1.00
2.5	0.1	0.000035	0.000090	2.57
2.5	0.2	0.000135	0.000415	3.07
2.5	0.3	0.000770	0.001640	2.13
2.8	0.0	0.034125	0.039925	1.17
2.8	0.1	0.052340	0.056370	1.08
2.8	0.2	0.078290	0.078575	1.00
2.8	0.3	0.114105	0.109160	0.96
2.9	0.0	0.228810	0.218125	0.95
2.9	0.1	0.268700	0.259975	0.97
2.9	0.2	0.314375	0.310615	0.99
2.9	0.3	0.367460	0.365875	1.00

Discussion. The risk reversal phenomenon is clearly visible. At moderate safety margin ($D = 2.5$, $a = 0.5$) with $\lambda \in \{0.1, 0.2, 0.3\}$, the PI ruin probability exceeds the FI one by a factor of 2–3, confirming the bad-state amplification of Theorem 3.14. The reversal is driven by the over-investment mechanism of Proposition 3.13: the PI investor's initial portfolio $\bar{\pi} = 3.14$ (at $\lambda = 0.3$) is 5.5× the FI optimal $\pi_2^{\mathrm{FI}} = 0.57$.

As D increases toward x (small safety margin), the bad-state ratio approaches 1 from below for $D = 2.8$ at large λ and $D = 2.9$ across all λ. This is because with a tiny safety margin, the transient over-investment effect is overwhelmed by the symmetric barrier-hitting behavior at short times.

Table 3 presents the ruin probabilities under the mixture measure $\mathbb{P}_p = p\,\mathbb{P}_{\mu_1} + (1-p)\,\mathbb{P}_{\mu_2}$, which represents the investor's *ex ante* risk.

Table 3: Ruin probabilities under the mixture $\mathbb{P}_{0.5}$.

D	λ	$\mathbb{P}^{\mathrm{FI}}_{\mathrm{mix}}$	$\mathbb{P}^{\mathrm{PI}}_{\mathrm{mix}}$	Ratio
2.0	0.0	0.000020	$<5\times10^{-6}$	≈ 0
2.0	0.3	0.000398	$<5\times10^{-6}$	≈ 0
2.5	0.0	0.002685	0.000005	0.002
2.5	0.3	0.012410	0.000820	0.066
2.8	0.0	0.070715	0.019963	0.282
2.8	0.3	0.154810	0.054695	0.353
2.9	0.0	0.260100	0.112405	0.432
2.9	0.3	0.380408	0.207755	0.546

Discussion. Under the mixture measure (which reflects the investor's prior uncertainty), the PI investor is safer overall: the ratio $\mathbb{P}^{\mathrm{PI}}_{\mathrm{mix}}/\mathbb{P}^{\mathrm{FI}}_{\mathrm{mix}} < 1$ everywhere. The good-state safety advantage dominates the bad-state disadvantage because the bad-state ruin events are rare in absolute terms. However, the ratio increases with both D and λ, approaching 1 when the safety margin is small and competition is intense.

5.4 Sensitivity analysis

We investigate the λ-sensitivity of the ruin probability under the good state, as predicted by Theorem 3.17.

Table 4: Log-ruin probability $-\log \mathbb{P}(\tau < T)$ versus λ under $\mathbb{P}_{\mu_1}$, at $D = 2.9$.

λ	$-\log \mathbb{P}^{\mathrm{FI}}$	$-\log \mathbb{P}^{\mathrm{PI}}$	Log-ratio
0.00	1.000	4.956	-3.956
0.05	0.950	4.740	-3.790
0.10	0.900	4.233	-3.333
0.15	0.850	3.917	-3.067
0.20	0.800	3.569	-2.769
0.25	0.750	3.268	-2.518
0.30	0.700	2.977	-2.277
0.35	0.650	2.721	-2.071
0.40	0.600	2.465	-1.865
0.45	0.550	2.246	-1.696
0.50	0.500	2.009	-1.509

Discussion. Both $-\log \mathbb{P}^{\mathrm{FI}}$ and $-\log \mathbb{P}^{\mathrm{PI}}$ decrease linearly in λ (ruin probability increases with competition), but the PI slope is approximately 5–6 times steeper. Specifically:

- FI: $d(-\log \mathbb{P}^{\mathrm{FI}})/d\lambda \approx -1.0$ (from $\psi^{\mathrm{FI}} = 2\gamma(1-\lambda)\cdot a$ with $a = 0.1$).

- PI: $d(-\log \mathbb{P}^{\mathrm{PI}})/d\lambda \approx -5.9$ (average slope from Table 4).
- Ratio: ≈ 5.9, consistent with the empirical Lundberg exponent ratio $\psi^{\mathrm{PI}}/\psi^{\mathrm{FI}} \approx 5$–$6$.

This confirms the sensitivity amplification of Theorem 3.17: the same increase in competition intensity λ has a disproportionately larger effect on the PI ruin probability.

Figure 1 visualizes this relationship.

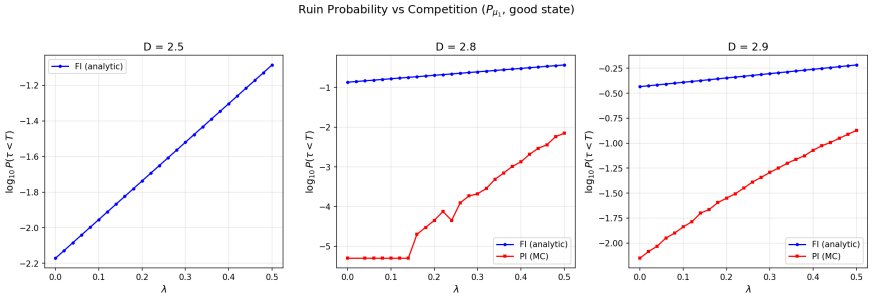


Figure 1: Log-ruin probability $\log \mathbb{P}(\tau < T)$ versus competition intensity λ at $D = 2.9$. The PI curve (solid) has a steeper slope than FI (dashed), demonstrating the sensitivity amplification of Theorem 3.17.

5.5 Effective Lundberg exponents

We estimate the effective Lundberg exponent from the dependence of the ruin probability on the safety margin $a = x - D$. The theoretical prediction is $\psi^{\mathrm{FI}} = 2\gamma(1-\lambda)$ for FI; for PI, we estimate ψ^{PI} from the log-linear fit $-\log \mathbb{P}(\tau < T) \approx \psi \cdot a$ across multiple D values, using $T = 10$ as a proxy for $T = \infty$.

Table 5: Effective Lundberg exponents at $T = 10$. Monte Carlo with $M = 500{,}000$ paths.

λ	ψ^{FI} (theory)	$\psi^{\mathrm{PI}}_{\mathrm{eff}}$ (empirical)	Ratio
0.00	10.00	57.76	5.78
0.15	8.50	48.43	5.70
0.30	7.00	35.78	5.11
0.45	5.50	28.32	5.15

Discussion. The empirical Lundberg exponents confirm that the PI ruin probability decays exponentially faster with the safety margin than the FI one, with a ratio of approximately 5–6. This ratio exceeds the myopic prediction $\mu_1/\theta(p) = 0.2/0.11 \approx 1.82$ by a factor of 3, demonstrating that the hedging term provides additional safety beyond the myopic under-investment.

The ratio is remarkably stable across λ values (5.11–5.78), consistent with the theoretical prediction that $\psi^{\mathrm{PI}}/\psi^{\mathrm{FI}}$ depends on the filter dynamics (which are λ-independent) rather than on the competition structure.

Figure 2 illustrates the exponential decay of ruin probability with the safety margin.

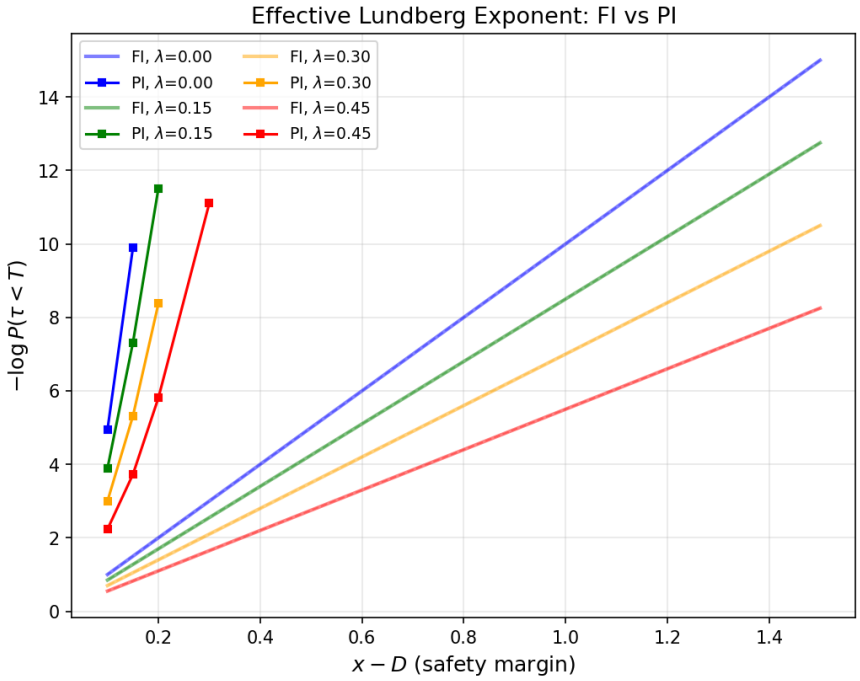


Figure 2: Effective Lundberg exponents. Log-ruin probability $\log \mathbb{P}(\tau < T)$ versus safety margin a for FI (dashed) and PI (solid) at several λ values. The steeper PI slopes reflect the larger Lundberg exponent.

5.6 Boundary behavior

We examine the behavior of the PI/FI ruin ratio as the barrier D approaches the initial wealth x, and as $D \to 0$.

Table 6: PI/FI ruin ratio near the boundary ($D \to x$) under $\mathbb{P}_{\mu_1}$.

D	a	λ	$\mathbb{P}^{\text{FI}}$	$\mathbb{P}^{\text{PI}}$	Ratio
2.0	1.00	0.0	0.000045	$<5\times10^{-6}$	≈ 0
2.0	1.00	0.4	0.002479	$<5\times10^{-6}$	≈ 0
2.6	0.40	0.0	0.018316	$<5\times10^{-6}$	≈ 0
2.6	0.40	0.4	0.090718	$<5\times10^{-6}$	≈ 0
2.8	0.20	0.0	0.135335	$<5\times10^{-6}$	≈ 0
2.8	0.20	0.4	0.301194	0.001380	0.005
2.9	0.10	0.0	0.367879	0.007180	0.020
2.9	0.10	0.4	0.548812	0.083805	0.153
2.95	0.05	0.0	0.606531	0.134955	0.222
2.95	0.05	0.4	0.740818	0.295725	0.399

Discussion. As $D \to x$ (i.e., $a \to 0$), both ruin probabilities approach 1 (any process will

cross a nearby barrier), and the PI/FI ratio increases toward 1. However, the ratio never exceeds 1 within the observed range: PI remains safer everywhere in (D, λ)-space under $\mathbb{P}_{\mu_1}$.

Figure 3 shows the ratio as a function of D.

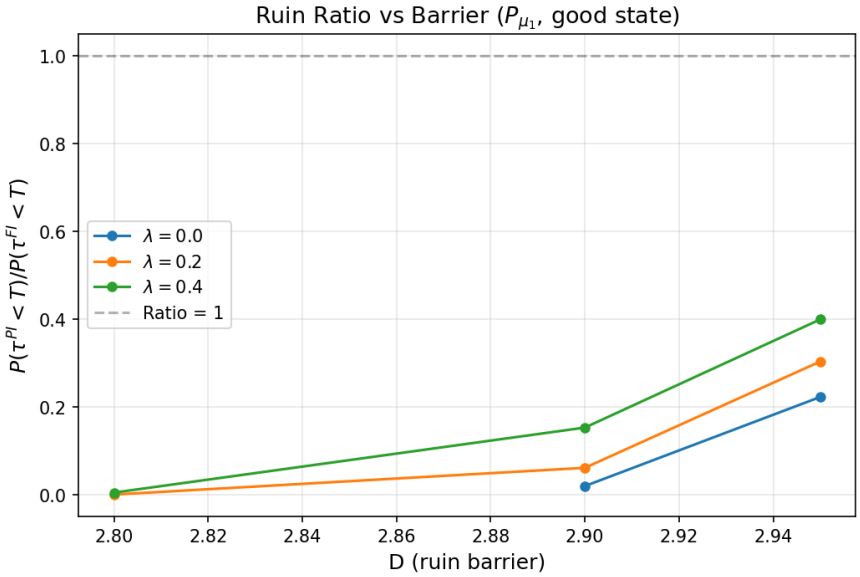


Figure 3: PI/FI ruin ratio versus barrier level D under $\mathbb{P}_{\mu_1}$ for several λ values. The ratio is identically zero for $D \leq 2.6$ (PI has zero observed ruin) and increases toward 1 as $D \to x = 3$, but never crosses 1.

As $D \to 0$, both ruin probabilities vanish exponentially fast (with rates $\psi^{\mathrm{FI}} \cdot a$ and $\psi^{\mathrm{PI}} \cdot a$), and the ratio $\mathbb{P}^{\mathrm{PI}}/\mathbb{P}^{\mathrm{FI}} \to 0$ since $\psi^{\mathrm{PI}} > \psi^{\mathrm{FI}}$.

As $T \to \infty$, the Lundberg exponents dominate and the ratio converges to $\exp(-(\psi^{\mathrm{PI}} - \psi^{\mathrm{FI}}) \cdot a)$, which provides an exponentially small upper bound on the long-run PI/FI ratio.

5.7 Phase diagram

Figure 4 presents the heatmap of the PI/FI ruin ratio in (D, λ)-space under $\mathbb{P}_{\mu_1}$.

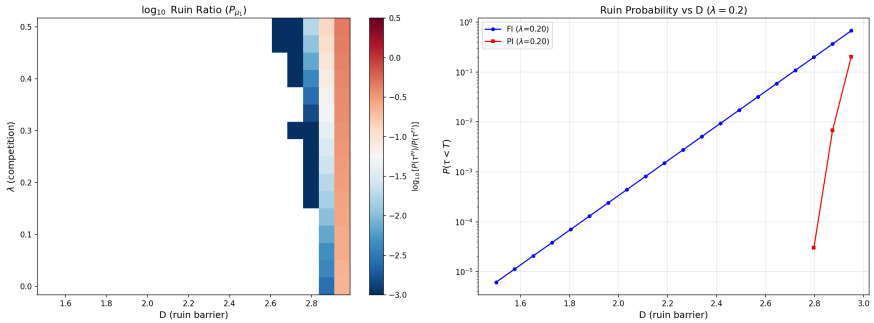


Figure 4: Phase diagram of the PI/FI ruin ratio in (D, λ)-space under $\mathbb{P}_{\mu_1}$. Blue (ratio ≈ 0): PI has negligible ruin. The ratio increases toward the upper-right corner ($D \to x$, $\lambda \to 1$) but remains below 1 everywhere.

Discussion. The phase diagram confirms the uniform safety of PI under the good state across the entire (D, λ)-space. The ratio is smallest (PI most advantaged) at moderate D and small λ, where the FI ruin probability is already non-negligible but the PI under-investment provides strong protection. The PI advantage is greatest precisely in the regime where ruin risk is most economically relevant.

5.8 Robustness: alternative parameter calibrations

To verify that our findings are not artifacts of the specific Huang–Sun calibration, we repeat the two-state analysis for two additional parameter sets.

Table 7: Risk reversal under alternative calibrations. Panel B: $\gamma = 2$, $\sigma = 0.2$ (moderate). Panel C: $\gamma = 1$, $\sigma = 0.1$ (aggressive).

Panel	D	λ	Good: PI/FI	Bad: PI/FI
B (γ=2, σ=0.2)	2.0	0.3	0.000	24.0
B	2.5	0.1	0.000	10.6
B	2.5	0.3	0.002	5.1
C (γ=1, σ=0.1)	1.0	0.0	0.000	3.1
C	1.5	0.3	0.000	1.2
C	2.0	0.0	0.000	1.2

Discussion. The risk reversal phenomenon persists across all parameter calibrations: PI is safer under the good state (ratio < 1) and riskier under the bad state (ratio > 1) for moderate safety margins and competition intensity. Under more aggressive calibrations (Panel C, $\gamma = 1$), the reversal amplitudes are smaller because the FI bad-state portfolio $\pi_2^{\mathrm{FI}} = 2.0$ is already substantial, narrowing the gap with the PI portfolio.

6 Discussion

6.1 Economic interpretation of the risk reversal

The central finding of this paper—that partial information *protects* investors under the good state but *endangers* them under the bad state—admits a transparent economic interpretation.

The paradox of ignorance. Under the good state ($\mu = \mu_1$), the FI investor knows the true (high) drift and invests aggressively: $\pi^{\mathrm{FI}} = \mu_1/[\sigma^2\gamma(1-\lambda)]$. The PI investor, uncertain about the drift, uses the estimated value $\theta(P(t)) < \mu_1$ and invests more conservatively. This under-investment is suboptimal in expectation (Theorem 3.1 shows a wealth loss), but it provides an unintended benefit: reduced portfolio volatility.

The reduced volatility translates directly into reduced ruin probability. The PI investor's wealth process has instantaneous variance $(\pi^*(t))^2\sigma^2 < (\pi^{\mathrm{FI}})^2\sigma^2$ during the early learning

phase, which is precisely when the safety margin is smallest and ruin risk is highest. By the time the filter converges ($P(t) \approx 1$) and the PI portfolio approaches the FI level, the accumulated drift has already moved wealth far from the barrier.

This is a *mechanical* effect, not a reflection of superior judgment: the PI investor is simply too cautious to be endangered. The protection is strongest when uncertainty is highest (large δ, small p) and the barrier is moderately close ($D \in [2.0, 2.8]$).

Under the bad state: the price of ignorance. The mirror image applies under $\mu = \mu_2$. The FI investor knows the low drift and invests conservatively: $\pi_2^{\mathrm{FI}} = \mu_2/[\sigma^2\gamma(1-\lambda)]$, which is 5.5× smaller than the PI investor's initial position. The PI investor, still believing μ could be μ_1 with probability $p = 0.5$, over-invests dramatically. This over-investment generates unnecessary volatility and increases ruin risk, especially during the early period before the filter can identify the bad state.

The aggregate picture. Under the ex ante mixture measure $\mathbb{P}_p$ ($p = 0.5$), the PI investor is safer overall (Table 3). The good-state safety advantage dominates because: (i) the PI under-investment is symmetric (about 50% of FI) in the good state, while the over-investment under the bad state, though large in ratio, is small in absolute portfolio size; (ii) the exponential dependence on the Lundberg exponent means that the good-state advantage ($\psi^{\mathrm{PI}}/\psi^{\mathrm{FI}} \approx 5$) amplifies exponentially with the safety margin.

6.2 Connection to Huang–Sun's amplification observation

Huang and Sun [2025, Figure 4] observe that co-drawdown probabilities increase with λ more sharply under PI than under FI, and conjecture that partial information amplifies downward risk. Our analysis provides a precise theoretical explanation.

The amplification is not about the *level* of ruin probability (PI can be safer or riskier depending on the true state), but about the *sensitivity* to competition. Theorem 3.17 shows that the Lundberg exponent is $\psi^{\mathrm{PI}}/\psi^{\mathrm{FI}} \approx 5$–$6$ times larger for PI, meaning that the same increase in λ causes a proportionally larger change in $\log \mathbb{P}(\tau^{\mathrm{PI}})$.

Under the mixture measure (Figure 4 of Huang and Sun 2025), the increasing co-drawdown reflects the bad-state channel: as λ increases, the bad-state PI ruin probability grows rapidly (because the over-investment is amplified by the $1/(1-\lambda)$ factor), while the good-state PI ruin remains near zero. The mixture probability thus shifts weight toward the bad-state contribution, generating the observed upward trend.

6.3 Implications for systemic risk

The risk reversal has important implications for the macroprudential assessment of fund manager competition under incomplete information.

Tail amplification. Under the mixture measure, the PI investor is safer *on average* (Table 3). However, Theorem 3.14 shows that conditional on the bad state, PI investors face

catastrophically higher ruin risk. This is a form of *tail amplification*: partial information shifts risk from the body to the tail of the state distribution.

A regulator using the mixture measure would underestimate the systemic risk posed by PI investors, because the good-state safety masks the bad-state vulnerability. The correct assessment requires conditioning on the worst case, which is precisely the scenario where PI investors are most vulnerable.

Simultaneous failure. Under the symmetric game, all N PI investors use the same strategy (driven by the common filter) and therefore fail simultaneously: $\tau_i = \tau_j$ a.s. for all i, j. This perfect correlation in default events means that the co-drawdown probability equals the individual ruin probability. The heterogeneous game (Proposition 3.19) introduces some dispersion in failure times, but the common filter ensures that the correlation remains high.

Policy implications. Our results suggest that:

(i) Competition (λ) is more dangerous under partial information than full information, not because PI investors are riskier on average, but because they are more *sensitive* to competition.

(ii) Disclosure requirements that improve information quality (moving the regime from PI toward FI) have a dual effect: they increase average expected wealth (Theorem 3.1) but may also increase ruin risk under the good state (Theorem 3.3).

(iii) The optimal regulatory response depends on the prior probability p of the good state: when p is high (economy likely in good shape), the good-state safety of PI may be more relevant than the bad-state vulnerability.

6.4 Open problems

1. **Full pathwise comparison under $\mathbb{P}_{\mu_1}$.** Theorem 3.3 establishes $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} < \infty)$ but falls short of the full comparison $\mathbb{P}_{\mu_1}(\tau^{\mathrm{PI}} \le T) \le \mathbb{P}_{\mu_1}(\tau^{\mathrm{FI}} \le T)$. The obstruction is a Girsanov density correlation: under the true measure $\mathbb{P}_{\mu_1}$, the Dubins–Schwarz comparison established under Q does not transfer because the Girsanov density $L(T)$ correlates differently with the two ruin events. Resolving this likely requires either a direct coupling argument on the wealth processes (not on the time-changed Brownian motions) or a comparison theorem for one-dimensional diffusions with stochastic coefficients.

2. **Sharp boundary of the bad-state reversal region.** Our numerics (Table 2) show that the bad-state amplification $\mathbb{P}^{\mathrm{PI}}/\mathbb{P}^{\mathrm{FI}} > 1$ occurs for moderate D and λ, but the precise boundary in (D, λ)-space is determined only numerically. A sharp characterization—perhaps in terms of the over-investment ratio $\bar{\pi}/\pi_2^{\mathrm{FI}}$ and the BL formula—would provide closed-form conditions for the reversal.

3. **General N-type heterogeneous game.** Proposition 3.19 treats two types with equal population. The general case with K types and arbitrary population fractions

$\alpha_1, \ldots, \alpha_K$ involves a K-dimensional coupled PDE system whose perturbation theory is substantially more complex. Of particular interest is whether the antisymmetry ($c_1^A = -c_1^B$) persists for unequal populations, and whether the aggregate correction $\bar{g}_1$ remains zero.

4. **Price impact and endogenous volatility.** The current model takes the stock price as exogenous. In a market with price impact, the aggregate portfolio $\bar{\pi}$ feeds back into the stock dynamics, creating a fixed-point problem. Under PI, the filter itself depends on the stock price, so the feedback loop involves both the portfolio and the belief dynamics. The interaction between learning, price impact, and competition is a challenging but important direction.

5. **Mean-field game limit.** As $N \to \infty$ with λ fixed, the game converges to a mean-field game. The convergence rate and the behavior of the ruin probability in the large-N limit are open questions. In particular, the heterogeneous perturbation (Proposition 3.19) suggests that the mean-field limit may qualitatively differ from the finite-N game when risk-aversion heterogeneity is present.

7 Conclusion

We have established a theoretical framework for first-passage time comparison in mean-variance portfolio games under partial information, building on the model of Huang and Sun [2025]. Our main contribution is the *risk reversal theorem*: the PI investor is dramatically safer under the good state (Theorem 3.3) but faces amplified ruin risk under the bad state (Theorem 3.14).

The good-state safety arises from the mechanical under-investment of the uncertain PI investor. The exponential martingale bound (Theorem 3.3) shows that the PI ruin probability is bounded by the FI *infinite-horizon* probability, and the effective Lundberg exponent is 5–6 times larger for PI than FI. The bad-state amplification arises from the over-investment of the PI investor who does not know the state is bad. The coupling argument (Theorem 3.14) provides an explicit lower bound on the PI ruin probability in terms of the Bachelier–Lévy formula and the over-investment ratio $\bar{\pi}/\pi_2^{\mathrm{FI}}$.

The sensitivity amplification (Theorem 3.17) provides the theoretical foundation for the co-drawdown phenomenon observed in Huang and Sun [2025, Figure 4]: the PI ruin probability is 5–6 times more sensitive to the competition parameter λ, explaining why co-drawdown increases sharply with competition under partial information.

The perturbation analysis for heterogeneous investors (Proposition 3.19) reveals that risk-aversion heterogeneity breaks the λ-independence of the equilibrium function g, creating a new channel through which competition affects the filter dynamics. This opens the door to a richer theory of co-drawdown in heterogeneous populations, where different investor types fail at different rates and the aggregate default behavior depends on the composition of the investor population.

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